



Detection of climate change signals using precipitation and temperature time series by a hybrid deep learning framework

Muhammad Waqas · Usa Humphries Wannasingha ·
Angkool Wangwongchai

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Abstract Climate change is one of the most extreme challenges of the twenty-first century. Precipitation (pr) and temperature variability are key indicators of climate change detection. Whereas hybrid deep learning (DL) models have been widely applied, their integration with region-specific datasets remains limited. This study uses a CNN-BiLSTM-TCN-attention model to identify the signals of climate change in Thailand based on the 1993–2024 station measurements and CMIP6-GCM simulations. Historical temperature analysis revealed statistically significant warming trends: coastal stations experienced T_{asmin} increases of up to 0.0673 °C/year and T_{asmax} of up to 0.0838 °C/year, urban areas 0.0401–0.0733 °C/

year, and high-altitude sites 0.02–0.03 °C/year. Precipitation trends were spatially heterogeneous, with increases at Khlong Yai (4.764 mm/year) and Samut Prakan (4.303 mm/year), but declines at Aranyaprathet (−0.688 mm/year) and Kampaeng Phet (−0.402 mm/year). After bias correction, the framework achieved high performance ($R^2=0.9807$ for T_{asmin} , 0.9782 for T_{asmax} , 0.9034 for precipitation) and low error metrics (MSE=0.0461). Future projections under SSP3-7.0 and SSP5-8.5 indicate widespread emergence of temperature signals, with median detection around 2019 and warming rates of +0.5 °C/decade; by 2100, T_{asmin} and T_{asmax} are projected to rise robustly (+0.50 to +0.72 °C/decade), while precipitation anomalies remain weakly negative (−0.03 to −0.08%/decade) with high variability but non-emergent trends. This study's findings show the importance of applying Thailand-specific observational and model data to effectively detect and quantify local climate change signals, which can be used to inform local adaptation planning.

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M. Waqas · U. H. Wannasingha · A. Wangwongchai (✉)
Department of Mathematics, Faculty of Science, King
Mongkut's University of Technology Thonburi (KMUTT),
Bangkok 10140, Thailand
e-mail: angkool.wan@kmutt.ac.th

M. Waqas
e-mail: muhammad.waqas@kmutt.ac.th

U. H. Wannasingha
e-mail: usa.wan@kmutt.ac.th

M. Waqas
The Joint Graduate School of Energy and Environment
(JGSEE), King Mongkut's University of Technology
Thonburi (KMUTT), Bangkok 10140, Thailand

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Introduction

Climate change and global warming are critical issues of the twenty-first century with immense

ramifications on natural ecosystems, socio-economic systems, and human lives (Fischer et al., 2005; Hariram et al., 2023). Surface temperature and precipitation (pr) are the most sensitive and easily measurable variables to detect climate change (Zhang et al., 2011b). These climate variables are the key drivers for evaluating long-term climatic trends and detecting climate change signals (Waqas et al., 2024c; Zhang et al., 2011b). Detection of such signals, statistically significant and persistent deviations of climatic variables outside the limits of natural variability, is critical to developing impact assessments, adaptation plans, and national climate action plans (Change, 2019; Ming et al., 2021).

To a large degree, the ability to simulate past and future climatic conditions is due to the Global Climate Models (GCM) framework, which is a part of the Coupled Model Intercomparison Project (CMIP) organized by the World Climate Research Program (WCRP) (Eyring et al., 2016; Luo et al., 2022). The CMIP6 suite provides multi-model ensembles globally, which are applied to Intergovernmental Panel on Climate Change (IPCC) assessments. However, GCMs have rough spatial resolution (250–600 km) and cannot be used on regional and local scales, particularly in a topographically and climatically diverse region such as Thailand (Humphries et al., 2024). To bridge this gap, methods of downscaling (both dynamical and statistical) have been proposed. Dynamic downscaling of Regional Climate Models (RCMs) is possible, i.e., simulating higher resolution by nesting finer-scale physics within GCM boundary conditions (Shah et al., 2024; Wood et al., 2004). Nevertheless, it remains a computationally demanding approach subject to parent GCM biases and susceptible to boundary instabilities (Xu et al., 2019). In its turn, statistical downscaling tries to reflect the empirical links between the large-scale predictors and the local-scale climate variables (Hellström & Chen, 2003; Salvi et al., 2011). Examples of conventional SD methods include linear regression (Montgomery et al., 2021; Tranmer & Elliot, 2008), delta change methods (Dang et al., 2020), and quantile mapping (Trinh-Tuan et al., 2019), and they tend to make linear and stationary assumptions, which are often not relevant to pr data with pronounced spatiotemporal variability and non-Gaussian distributions (Hadipour et al., 2016; Humphries et al., 2024).

Current progress in machine learning (ML) and deep learning (DL) can overcome the shortcomings of the classical models and trend detection techniques (Ladi et al., 2022). These models can develop complex non-linear connections, identify weak trends in extensive multidimensional data, and enhance the accuracy of predictions in variable weather conditions (Waqas et al., 2024c). In this case, CNNs and RNNs, along with LSTM-based architecture, have been quite helpful in extracting spatial and temporal features in climate data (Gilik et al., 2022; Kumar et al., 2020). When gridded data are provided, CNNs excel at learning spatial hierarchies and have been demonstrated to outperform in classifying spatial patterns of precipitation extremes and temperature anomalies (Chattopadhyay et al., 2020). The models of LSTM, which maintain long-term dependencies, are suitable for detecting temporal trends and multivariate time series forecasting (Rolnick et al., 2022). Combined, as, e.g., in hybrid CNN-LSTM models, they provide a unified fix to the dilemma of how to model spatial heterogeneity and time dynamics of climate systems (Reichstein et al., 2019; Yu, 2021). Also, DL models like autoencoders have been used more in unsupervised learning tasks, e.g., anomaly detection and dimensionality reduction (Waqas et al., 2024b). Regarding climate change, autoencoders help detect indirect, non-obvious changes in climatic patterns that are difficult to detect with standard statistical methods (Ibrahim et al., 2021). Notably, these DL models can now be interpreted using explainable AI (XAI) methods, which allow one to understand what physical processes drive the changes predicted by the model and enable it to be more realistic in decision-making (Samek et al., 2017, 2021).

Thailand is located in the center of the South-east Asian (SEA) monsoon system and is susceptible to climate change (Waqas et al., 2024d). Its wide range of agro-ecological zones and a high reliance on climate-sensitive sectors like agriculture and water resources make the country highly vulnerable to changes and fluctuations in pr and temperature (Waqas et al., 2024f).

Despite developments in ML/DL models, there remains a significant research gap in climate change detection in Thailand, where climate data can be scarce, inconsistent, or vary in quality (Waqas et al., 2023). Most current research applying AI to climate applications has focused on prediction (e.g.,

forecasting temperature or rainfall) or atmospheric greenhouse gas data based on satellite platforms, including GOSAT, OCO-2, or Copernicus (Buchwitz et al., 2018). Similarly, in another study, a DNN model was used for monthly rainfall prediction in the eastern region, based on the data of one station. This model showed a decent short-term predictive ability; approximately 70% of the forecasts aligned with observations, but the accuracy declined with longer lead times (Weesakul et al., 2018). Past studies highlight the increasing number of DL models in climate resilience studies, with neural networks, such as CNNs, RNNs, LSTMs, and deep belief networks, making up more than 22% of AI-based studies in the field (Ladi et al., 2022). Its application has also been used to examine climate extremes (e.g., floods, droughts, wildfires, heatwaves), where noise, sparsity, or heterogeneity of data are addressed with more transparent, interpretable, and stakeholder-trustworthy models (Camps-Valls et al., 2025). Future trends are causally informed DL frameworks that combine causal discovery techniques to increase model stability and reliability in climate applications (Iglesias-Suarez et al., 2024).

Most trend detection studies in Thailand have been based on single-source data (e.g., station-based observations) and linear methods of analysis (Masud et al., 2016; Reda et al., 2013). These methods do not consider the possible advantages of data fusion and the new signal extraction algorithms. The combination of various data sources, such as ground stations, satellite products, CMIP6 GCMs, or reanalysis datasets, into a DL framework can significantly improve the resilience and spatial extent of trend studies (Waqas & Humphries, 2024; Waqas et al., 2024c).

Integrating a DL framework capable of capturing spatial, temporal, and non-linear relationships facilitates robust detection of climate change signals. When applied to Thailand-specific observational data and validated across diverse agro-ecological regions, such models improve the identification of localized climate trends. This study addresses a key research gap: distinguishing between predicting and detecting climate change signals. The objectives are: (1) to detect and characterize long-term climate change signals in historical precipitation and temperature time series across Thailand, and (2) to apply and validate a hybrid DL framework that combines station-based observations with CMIP6-GCM outputs to identify

non-linear and spatially explicit shifts in climatic trends. This approach enhances understanding of the spatiotemporal dynamics of climate change in Thailand.

Materials and methods

Study area

This study focused on Thailand, a tropical country in SEA located between latitudes 5°–20°N and longitudes 97°–106°E. The highland and mountainous terrain of the northern part has a tropical climate, with wet and dry seasons, moderate to high altitude, and complex orographic effects on the local precipitation and temperature gradients (Waqas et al., 2024f). The Northeastern Korat Plateau has a semi-arid monsoonal climate with marked seasonality in rainfall and recurrent droughts that severely impact rainfed agricultural production (Waqas et al., 2024a). Central Thailand is a region of large floodplain systems of alluvial soil, which has a tropical wet and dry climate that favors intensive rice production but is highly prone to seasonal floods and waterlogged conditions (Waqas et al., 2024e). Conversely, the Southern Peninsula is characterized by a humid tropical climate characterized by high annual rainfall, high humidity, and a small range of seasonal temperature variation, and is more suited to perennial crops and dense forest systems (Humphries et al., 2024). The heterogeneous agro-ecological potential of the region is based on these spatial climatic gradients, as well as the presence of a wide range of soil types and elevations (Waqas et al., 2024c).

Data description and sources

Observation data

Monthly meteorological data between 1993 and 2024 were acquired in 129 Thai Meteorological Department (TMD) stations, including p_r , maximum temperature (T_{asmax}), and minimum temperature (T_{asmin}). Strict quality control measures were used, including missing values detection and imputations, homogeneity testing, and outlier analysis. Three stations with more than 50% missing records were omitted from the study to achieve dataset reliability (Table 1). All

Table 1 Monthly descriptive statistics of the observation dataset from TMD from 1993 to 2024

Month	T_{asmin}			T_{asmax}			Pr		
	Mean	Min	Max	Mean	Min	Max	Mean	Sum	Max
January	19.23	9.69	27.04	31.18	18.27	35.66	40.00	151,195.85	1807.93
February	20.56	9.30	27.88	32.93	20.54	38.01	38.04	143,805.10	617.00
March	22.83	14.30	37.82	34.61	21.54	40.50	84.54	319,543.97	1543.70
April	24.43	15.20	66.66	35.41	24.48	41.96	118.38	447,463.22	630.90
May	24.87	0.39	29.28	34.25	22.66	40.26	176.60	667,552.97	1043.10
June	24.81	0.00	28.47	33.25	22.51	38.31	190.63	720,597.76	1121.74
July	24.55	17.24	28.38	32.47	21.92	37.10	213.27	806,166.14	2106.20
August	24.40	17.17	28.31	32.17	21.57	35.79	214.52	810,872.03	1707.20
September	24.07	16.73	31.00	32.03	21.49	35.75	213.26	806,136.40	1920.80
October	23.32	12.37	69.59	31.81	20.25	36.10	154.67	584,650.76	1156.70
November	21.75	12.92	27.61	31.46	17.93	35.37	100.95	381,575.42	1397.80
December	19.57	10.20	27.04	30.42	15.17	35.21	66.76	252,347.07	1324.60

retained stations were subjected to comprehensive and station-wise descriptive statistics (Table S1).

CMIP6-GCMs dataset

The research uses a set of CMIP6 GCMs as shown in Table 2. ACCESS-CM2, MIROC6, and MPI-ESM variants are robust models of pr and T_{asmin} , T_{asmax} , whereas CMCC-ESM2 integrates biogeochemical cycles to assess long-term climate. High-resolution models such as MPI-ESM1-2-HR have the advantage of better atmospheric–ocean dynamics, but lower-resolution versions such as MPI-ESM1-2-LR are computationally more efficient. The historical and SSP370/SSP585 scenarios comprehensively assess historical climate variability, future projections, and climate change detection.

Methods

TCN's model for historical climate signal detection

Detecting long-term climate change signals within historical pr, T_{asmin} , and T_{asmax} time series requires more advanced computational procedures that can identify both abrupt and gradual changes as well as high-order, non-linear associations in spatiotemporal data.

Non-parametric methods such as Mann–Kendall (MK) trend test (Kendall & Stuart, 1977) and slope estimator of Sen (Hamed & Rao, 1998) have been commonly used to analyze climatic time series due to

their resistance to non-normality and outliers (Waqas et al., 2024c). However, these approaches are limited because they are monotonic and stationary, and they do not tend to capture complex temporal dependencies, higher-order interactions, and multi-scale variability common in climatic processes (Hamed, 2009; Waqas et al., 2024c).

To overcome these limitations, the study employs TCNs, a DL model designed to process sequential data that combines the representational power of CNNs with architectural modifications that preserve the temporal causality and allow for extracting long-range dependencies (Bai et al., 2018).

Data preprocessing

The workflow begins with retrieving the TMD's station-based pr and temperature data, supplemented with ERA5 reanalysis datasets for spatial completeness and temporal continuity. The data often contain missing entries due to observational gaps.

Let $P(s_i, t)$ denote the pr at the station s_i and time t , and $T_{min}(s_i, t)$, $T_{max}(s_i, t)$ represent the corresponding T_{asmin} and T_{asmax} records. Such raw data usually has missing data because of an observational gap. Missing values are imputed using an optimized k-nearest neighbors (k-NN) algorithm with $k=5$, where weights are assigned as the inverse Euclidean distance of the neighbors, ensuring closer neighbors have a greater influence (Wangwongchai et al., 2023). It calculates the estimate of a missing observation as a weighted average of the k nearest neighbors of that

Table 2 Selected CMIP6 GCM models, their sources, descriptions, and resolution

Model	Institution/source	Experiments	Variables	Members	Description
ACCESS-CM2	CSIRO-ARCCSS (Australia)	Historical, ssp370, ssp585	pr, T_{asmax} , T_{asmin}	r1i1p1f1	Coupled atmosphere–ocean GCM, widely used in CMIP6, is suitable for regional climate projections over the SEA
AWI-CM-1–1-MR	Alfred Wegener Institute (Germany)	Historical, ssp370, ssp585	pr, T_{asmax} , T_{asmin}	r1i1p1f1	Medium-resolution climate model emphasizing ocean–atmosphere interactions; often used in sensitivity studies
BCC-CSM2-MR	Beijing Climate Center/NUIST (China)	Historical, ssp370, ssp585	pr, T_{asmax} , T_{asmin}	r1i1p1f1	A fully coupled Earth system model that includes the carbon cycle and aerosol feedback, applied in monsoon studies
CMCC-ESM2	CMCC (Italy)	Historical, ssp370, ssp585	pr, T_{asmax} , T_{asmin}	r1i1p1f1	The Earth System Model version includes biogeochemical cycles and is used for long-term climate projections
FGOALS-g3	Chinese Academy of Sciences (CAMS)	Historical, ssp370, ssp585	pr, T_{asmax} , T_{asmin}	r1i1p1f1	Global coupled climate models are widely applied in Asia’s CMIP6 projections and temperature studies
MIROC6	MIROC Consortium (Japan)	Historical, ssp370, ssp585	pr, T_{asmax} , T_{asmin}	r1i1p1f1	Coupled climate model with ocean and atmosphere components; commonly used in East Asia climate studies
MPI-ESM1-2-HR	Max Planck Institute for Meteorology (Germany)	Historical, ssp370, ssp585	pr, T_{asmax} , T_{asmin}	r1i1p1f1	High-resolution spectral model; resolves detailed atmospheric and oceanic processes for CMIP6 scenarios
MPI-ESM1-2-LR	Max Planck Institute for Meteorology (Germany)	Historical, ssp370, ssp585	pr, T_{asmax} , T_{asmin}	r1i1p1f1	Low-resolution variant; computationally efficient for long-term simulations and scenario projections
MRI-ESM2-0	Meteorological Research Institute (Japan)	Historical, ssp370, ssp585	pr, T_{asmax} , T_{asmin}	r1i1p1f1	Coupled Earth system model; includes detailed land surface and atmospheric chemistry processes

observation $x_{i,t}$ as the weighted means of its k most similar historical instances in the feature space:

$$x_{i,t} = \frac{\sum_{j=1}^k w_j * x_{j,t}}{\sum_{j=1}^k w_j}, \quad (1)$$

where $w_j = \frac{1}{\text{dist}(x_{i,t}, x_{j,t})}$ is the inverse Euclidean distance weight between the target record and its j^{th} neighbor, ensuring that closer instances in the feature space exert greater influence on the imputation.

After gap-filling, the data are aggregated from daily or monthly resolutions to seasonal and annual scales to reduce high-frequency noise while preserving low-frequency climate variability. The processed dataset can be represented as a tensor $X \in \mathbb{R}^{N \times T \times D}$, where N is the number of stations, T is the number of time steps, and D is the number of climatic variables.

TCN architecture

TCNs are advantageous over RNNs (LSTM, GRU) due to parallelizable convolutions, avoidance of vanishing/exploding gradients, and long-range dependency capture through dilated causal convolutions (Bai et al., 2018; Lea et al., 2017). The trick to TCNs lies in causal convolutions, which predict time dependence based solely on inputs at time t and earlier, thereby preserving temporal order.

For a univariate timeseries x_t , the dilated convolution at layer l with kernel K_l of size k and dilation factor d is defined as:

$$(K_{l \otimes d} x) = \sum_{i=0}^{k-1} K_l(i) \times x_{t-d \cdot i} \quad (2)$$

Here, d controls the spacing between elements in the input over which the convolution is applied. By increasing exponentially with layer depth according to $d = 2^{l-1}$, the receptive field of the network expands exponentially, enabling the capture of dependencies over hundreds or thousands of time steps without requiring proportionally large kernels or excessive network depth.

Every residual block in the TCN structure has two dilated causal convolution layers with weight normalization (Salimans & Kingma, 2016) and rectified linear unit (ReLU) activation applied to them (Salimans & Kingma, 2016). It is a reparameterization of the convolutional kernel by weight normalization. K_l as:

$$K_l = g \cdot \frac{v}{\|v\|} \quad (3)$$

where v is the unnormalized weight vector, g is a learnable scalar, and $\|v\|$ denotes the Euclidean norm. This formulation separates the length and direction of

the weight vector and increases training stability and convergence. A skip connection between the block's input and output is introduced to alleviate the vanishing gradient problem and enable the network to learn identity mappings when it helps. The output of the residual block is therefore:

$$o_l = \text{ReLU}(o_{l-1} + F(o_{l-1})) \quad (4)$$

where $F(\cdot)$ denotes sequential dilated convolution, weight normalization, and activation functions. The network is trained to optimize a combination loss, a trade-off between trend fidelity and sensitivity to sudden changes in the time series. In particular, the loss function L is given by:

$$L = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2 + \beta \lambda \sum_{t=2}^T \mathbb{I}(|\hat{y}_t - \hat{y}_{t-1}| > \tau) \cdot (\hat{y}_t - \hat{y}_{t-1})^2 \quad (5)$$

where y_t and $\hat{y}_t$ are observed and predicted values, respectively. λ controls the weight of the abrupt change penalty, where $\tau = 1$ mm/day for pr and 0.5 °C for temperatures (empirical thresholds), and the Adam optimizer is applied with a learning rate of 0.001.

Model validation

A spatiotemporal cross-validation scheme is applied to rigorously assess the model's performance, in which the data are divided into K geographically stratified folds. In every iteration, $K-1$ folds will be used in training and the other in testing, so that evaluation is done on spatially separate regions to determine generalizability across the various climatic zones in Thailand. Nash Sutcliffe efficiency (NSE) is used to quantify model performance.

$$NSE = 1 - \frac{\sum_{t=1}^T (y_t - \hat{y}_t)^2}{\sum_{t=1}^T (y_t - \bar{y})^2} \quad (6)$$

where $\bar{y}$ is the mean of observed values. Also, the MK statistic is calculated on the actual series, and the modified MK statistic (Hamed & Rao, 1998; Waqas et al., 2024c) is calculated on the predicted series to see whether identified trends are consistent with the historical trends.

Spatial distribution of trends detected is produced after prediction by combining TCN output with geostatistical interpolation. Specifically, the TCN-derived trend magnitude $\beta(s_i)$ at station s_i is interpolated to unsampled locations s_0 via ordinary kriging:

$$\hat{\beta}(s_0) = \sum_{i=1}^N \lambda_i \beta(s_i) \quad (7)$$

The weights λ_i are determined from the variogram $\gamma(h)$, which quantifies the spatial dependence between pairs of observations separated by distance h :

$$\gamma(h) = \frac{1}{2|N(h)|} \sum_{i,j \in N(h)} [\beta(s_i) - \beta(s_j)]^2 \quad (8)$$

where $N(h)$ denotes the set of station pairs at lag h . The resulting interpolated spatial fields of pr and temperature trends in Thailand can be used to detect climate change signals at a regional scale.

SHapley additive exPlanations for attribution

SHapley additive exPlanations (SHAP) (Lundberg, 2017) are employed to attribute detected changes to potential drivers. SHAP values decompose the TCN prediction $\hat{y}_t$ into additive contributions from each input feature j :

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{j\}) - f(S)] \quad (9)$$

where F is the set of all features and $f(S)$ is the model prediction using the subset S . It enables the separation of anthropogenic forcing (e.g., greenhouse gas concentrations) and natural variability events. By combining TCNs, geostatistical interpolation, and explainable AI, the approach enables robust detection, spatial mapping, and the identification of long-term climate change signals in Thai precipitation and temperature time series, thereby overcoming the limitations of both conventional statistical and conventional DL models.

CNN-BiLSTM-TCN-attention hybrid framework

To address spatiotemporal variability and non-linearity, this study integrates a CNN-BiLSTM-TCN-Attention hybrid model (Fig. 1), which captures spatial patterns and long-term temporal

dependencies. The choice of CNN, BiLSTM, and TCN modules is intended to learn fine-grained local dependencies and long-range temporal correlations concurrently and deal with the non-linearity inherent in climate time series.

Data preprocessing

The TMD daily and monthly station observations, historical reanalysis data, and CMIP6 GCM projections were used under SSP370 and SSP585 scenarios. The variables considered include pr, T_{asmin} , and T_{asmax} over multi-decadal periods.

For a CMIP6-GCMs time series $X = \{x_1, x_2, \dots, x_T\}$ with missing entries, the E-step computes the conditional expectation of the missing values given the observed values and current parameter estimates $\theta^{(k)} = (\mu^{(k)}, \Sigma^{(k)})$ as

$$\theta_{missing}^{(k+1)} = \mathbb{E}[x_{missing} | x_{observed}, \theta^{(k)}] \quad (10)$$

Moreover, the M-step updates the mean and covariance to maximize the expected complete-data log-likelihood. It is repeated until convergence. To minimize seasonal effects and enhance inter-station comparability, monthly z-score normalization was applied:

$$z_t = \frac{x_t - \mu_{month(t)}}{\sigma_{month(t)}}, \quad (11)$$

where $\mu_{month(t)}$ and $\sigma_{month(t)}$ are the long-term mean and standard deviation for the month corresponding to time index t .

Spatial alignment of station and GCM grids was achieved via bilinear interpolation, mapping irregular station coordinates to the regular grid of GCM data (Hu et al., 2019). For each station, the nearest GCM grid point was selected, and bias correction was applied as:

$$x_{corrected} = x_{obs} + (x_{obs} - x_{GCM}) \quad (12)$$

CNN module

Spatial features of the gridded climate information were extracted via CNN layers:

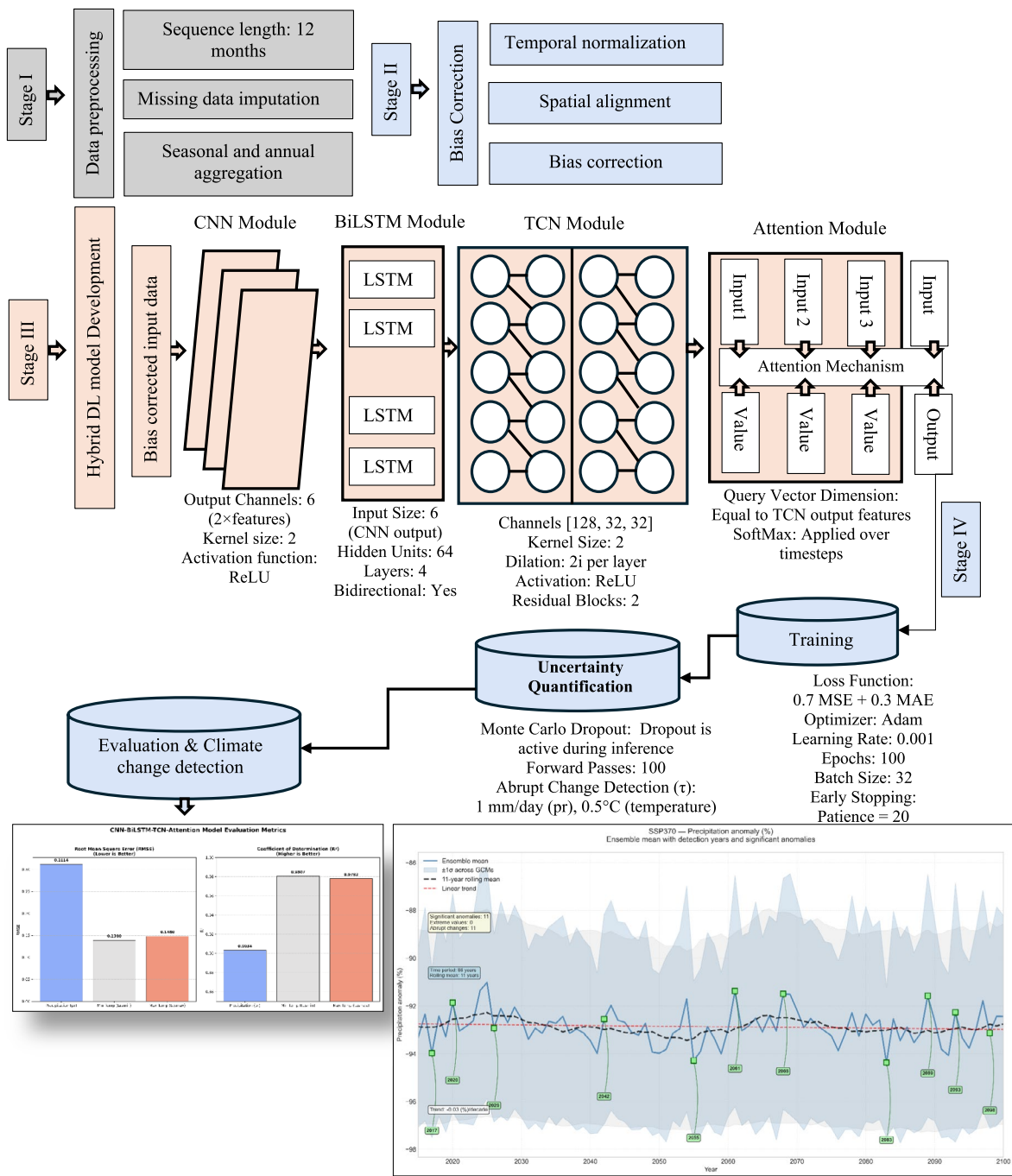


Fig. 1 Working architecture of CNN-BiLSTM-TCN-attention hybrid framework

$$Y_{i,j}^l = \sigma \cdot \left(\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} W_{m,n}^{(l)} \cdot X_{i+m,j+n}^{(l-1)} + b^{(l)} \right) \quad (13)$$

where $W_{m,n}^{(l)}$ denotes the learnable kernel weights of size $M \times N$, $b^{(l)}$ is the bias term, $\sigma(\cdot)$ is the ReLU activation function, and $X^{(l-1)}$ is the input feature map from the previous layer. The spatial resolution is

decreased through max-pooling layers, and the most salient features are maintained to regulate the overfitting and computational cost.

BiLSTM module

Although CNN layers work well in representing spatial dependencies, long-range temporal dependencies are needed to detect climate trends. It is dealt with using BiLSTM networks (Hochreiter, 1997). In an LSTM cell, the key gating mechanisms—input gate i_t , forget gate f_t , and output gate o_t —are computed as

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i), \quad (14)$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f), \quad (15)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o), \quad (16)$$

with the candidate cell state given by:

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c), \quad (17)$$

Moreover, the cell state is updated as:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad (18)$$

where $\odot$ denotes element-wise multiplication. The hidden state output is.

$$h_t = o_t \odot \tanh(c_t). \quad (19)$$

In the bidirectional setting, a forward LSTM processes the sequence from $t = 1$ to T and a backwards LSTM processes it in reverse, producing final hidden states.

$$h_t = \vec{h}_t \oplus \bar{h}_t, \quad (20)$$

where $\oplus$ denotes concatenation.

TCN module

As well as the BiLSTM, the architecture includes a TCN module, which applies 1D causal convolutions with dilations to produce a temporal dependency model without recurrent connections (Bai et al., 2018). The convolution of a sequence x with a filter f of length k and dilation factor d is a dilated convolution by

$$y(t) = \sum_{i=0}^{k-1} f(i) \cdot x(t - d \cdot i). \quad (21)$$

The design ensures that the output only relies on inputs at t and earlier times and is causal. The receptivity field grows exponentially with the number of layers and the dilation factors, which enables learning of long-range dependencies. The residual connections stabilize training and gradient flow. Compared to RNN-based models, TCNs are parallelizable in sequence length, which is computationally more efficient in dealing with big climate data (Bai et al., 2018; Lea et al., 2017). The BiLSTM-TCN model structure enables the model to learn long-term sequential dependencies and multi-scale temporal patterns, which are desirable for detecting weak and non-linear climate trends.

Attention module

It takes the output sequences of the temporal modules as input to increase the interpretability and flexibly balance the temporal features using a self-attention mechanism (Ashish, 2017). Given hidden states h_t , the attention weights are computed in the following way:

$$\alpha_t = \frac{\exp(q^T h_t)}{\sum_{j=1}^T \exp(q^T h_j)}. \quad (22)$$

where q is a learnable query vector, the resulting context vector is the weighted sum.

$$c = \sum_{t=1}^T \alpha_t h_t, \quad (23)$$

which is passed on to fully connected layers to make the final prediction.

Training and uncertainty quantification

The training is performed using a compound loss of mean squared error (MSE) and mean absolute error (MAE),

$$\mathcal{L} = \lambda \cdot \text{MSE}(y, \hat{y}) + (1 - \lambda) \cdot \text{MAE}(y, \hat{y}). \quad (24)$$

where.

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (25)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (26)$$

Furthermore, λ is empirically fixed as 0.7. The optimization is done with the Adam algorithm (Kingma, 2014), which updates parameters at each step.

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (27)$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1}, \hat{v}_t = \frac{v_t}{1 - \beta_2},$$

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}},$$

where g_t is the gradient at time t , and β_1 and β_2 are decay rates for the first and second moments. Overfitting is avoided by early stopping with a patience of 20 epochs.

The Monte Carlo dropout is used to quantify uncertainty (Gal & Ghahramani, 2016): dropout layers are kept active during inference, and predictions are stochastic. In the case of NNN stochastic forward passes, the predictive mean is.

$$\bar{y} = \frac{1}{N} \sum_{i=1}^N \hat{y}^{(i)}. \quad (28)$$

Moreover, the predictive variance is.

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (\hat{y}^{(i)} - \bar{y})^2. \quad (29)$$

Furthermore, Table 3 provides all component parameters, preprocessing choices, hyperparameters, and their reasoning, ensuring workflow reproducibility.

Sensitivity analysis and evaluation criteria

A sensitivity analysis was conducted to assess model robustness and the influence of key hyperparameters.

Model performance was evaluated under systematic perturbations of sequence length, BiLSTM hidden units, CNN kernel sizes, dropout rates, and learning rate. Additionally, input datasets were perturbed with Gaussian noise $\epsilon \sim N(0, \sigma_{obs}^2)$, where σ_{obs}^2 is the standard deviation of observational errors, and the resulting variation in predictive performance was quantified.

Formally, for an input perturbation $X' = X + \epsilon$, the sensitivity index S for each model hyperparameter or input feature was computed as follows:

$$S = \frac{Var(\hat{y}(X'))}{Var(\hat{y}(X))} \quad (30)$$

S values near 1 are used to show low sensitivity, and deviations are used to show parameters that significantly impact model predictions. It made the model less reliant on a particular hyperparameter configuration or individual observations, which is more likely to improve the generalizability (Saltelli et al., 2008).

The CNN-BiLSTM-TCN-Attention hybrid framework was evaluated by comparing its performance to that of statistical downscaling methods and other hybrid DL frameworks. Bias-corrected delta methods and quantile mapping were used as statistical methods (Maraun et al., 2010). CNN-LSTM and BiLSTM-TCN were used as DL comparisons.

Various measures were used to assess model performance, which included NSE, MAE, R^2 , RMSE, and KGE: MK statistic and trend consistency between observed and predicted trends.

Point metrics, uncertainty estimates, and 95% confidence intervals were reported to measure predictive skill and reliability comprehensively. The framework explicitly facilitates the environmental monitoring goals by integrating TCN-based trend detection, CNN-BiLSTM hybrid modeling, attention mechanisms, and intensive uncertainty quantification. Confidence interval projected trends enable spatially and temporally resolved evaluation of climate change effects, which can be acted upon to inform water resource planning, agricultural planning, and climate change adaptation planning. The comparison with other models has verified that the suggested solution enhances predictive quality and uncertainty description.

Table 3 CNN-BiLSTM-TCN-Attention hybrid framework parameters and hyperparameters

Component	Parameter/hyperparameter	Setting	Reasoning
Data and preprocessing	Sequence length	12 months	Captures one-year temporal patterns, ensuring seasonal dependencies are represented
	Features	3 (pr , T_{asmin} , T_{asmax})	Key climatic variables to capture precipitation and temperature variability
	Missing data imputation	k-NN ($k=5$) with inverse Euclidean distance weighting	Ensures realistic estimates for missing observations while giving higher weight to closer neighbors
	Aggregation	Seasonal & annual	Reduces high-frequency noise while preserving low-frequency climate variability
	Temporal normalization	Monthly z-score	Removes seasonal bias and ensures comparability across stations
	Spatial alignment	Bilinear interpolation to the GCM grid	Aligns irregular station coordinates with the regular GCM grid for consistency in spatial features
	Bias correction	Additive adjustment: $x_{corrected} = x_{obs} + (x_{obs} - x_{GCM})$	Corrects systematic biases between observed and modeled datasets
CNN	Input	3	Corresponds to pr , T_{asmin} , T_{asmax}
	Output channels	6 ($2 \times \text{features}$)	Expands feature representation for BiLSTM input
	Kernel size	2	Captures local spatial dependencies
	Activation	ReLU	Introduces non-linearity for better feature representation
	Pooling	Max-pooling	Reduces spatial resolution, prevents overfitting, and reduces computation
BiLSTM	Input size	6 (CNN output)	Receives spatial features for temporal modeling
	Hidden units	64	Balances model capacity and computational efficiency
	Layers	4	A simple structure is sufficient for medium-length sequences
	Bidirectional	Yes	Captures both forward and backward temporal dependencies
TCN	Channels	[128, 32, 32]	Gradually reduces feature dimensions while learning temporal patterns
	Kernel size	2	Captures short-term temporal patterns
	Dilation	2^i per layer	Expands receptive field exponentially to learn long-range dependencies
	Activation	ReLU	Non-linear transformation of temporal features
	Residual blocks	2 layers per block	Stabilizes training and mitigates vanishing gradient issues

Table 3 (continued)

Component	Parameter/hyperparameter	Setting	Reasoning
Attention	Query vector dimension	Equal to TCN output features	Learns importance weights across sequence timesteps
	SoftMax	Applied over timesteps	Normalizes attention weights
Fully connected output	Output size	3	Predicts pr , T_{asmin} , and T_{asmax} for the next time step
Training	Loss function	0.7 MSE + 0.3 MAE	Balances sensitivity to outliers (MAE) and overall fit (MSE)
	Optimizer	Adam	Adaptive learning rate, robust for complex architectures
	Learning Rate	0.001	Standard for Adam; adjustable if convergence issues occur
	Epochs	100	Sufficient for convergence with early stopping
	Batch size	32	Balances gradient stability and memory efficiency
	Early stopping	Patience = 20	Prevents overfitting
Uncertainty quantification	Monte Carlo dropout	Dropout is active during inference	Enables stochastic predictions for uncertainty estimation
	Forward passes	100	Computes predictive mean and variance for each prediction
Thresholds	Abrupt change detection (τ)	1 mm/day (pr), 0.5 °C (temperature)	Empirically selected to penalize extreme inter-annual differences in climate variables

Results

Historical climate detection via TCN framework

Temperature trends: warming patterns with regional variability

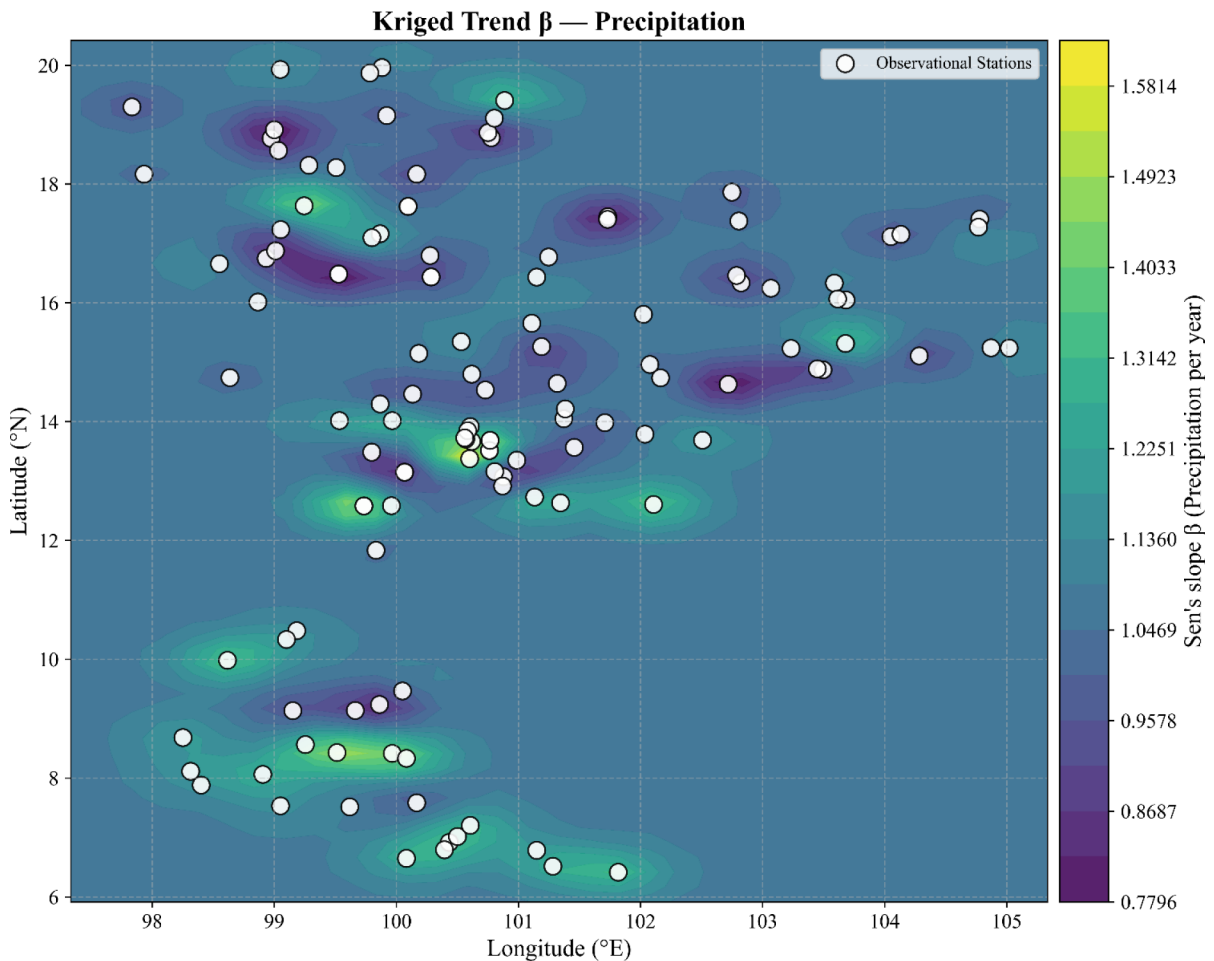
The assessment criteria, such as NSE, KGE, RMSE, and Bias, indicate that the TCN model can capture climate signals. The outcomes reveal non-homogeneous patterns nationwide, characterized by varying intensities and directions in T_{asmin} , T_{asmax} , and pr (Fig. 2 and Table S2).

TCN-derived coefficients of beta (β) of T_{asmin} and T_{asmax} indicate an overall warming trend in Thailand, but the degree is highly dependent on location (Fig. 2b and c). For example, Aranyaprathet has a T_{asmin} rise of 0.0221 °C/year and a T_{asmax} increase of 0.0331 °C/year, indicative of a more pronounced day warming signal. Even higher warming rates are observed at Ayutthaya Agro (0.0494 °C/year in T_{asmin} and 0.0490 °C/year in T_{asmax}), which may be associated with urban heat island effects, as it is located

close to Bangkok. Bang Khen Agro, close to Bangkok, is an exception, with slightly negative temperature trends (−0.0184 °C/year in T_{asmin} , −0.0147 °C/year in T_{asmax}), perhaps because of local land-use change or an observational bias.

The warming patterns align with global temperature records attributed to GHG forcing. Nevertheless, spatial heterogeneity implies that the larger climate signal is adjusted by regional parameters, e.g., by urbanization, deforestation, or irrigation. Coastal stations such as Chonburi (0.0438 °C/year T_{asmin} , 0.0623 °C/year T_{asmax}) and Krabi (0.0673 °C/year T_{asmin} , 0.0838 °C/year T_{asmax}) are warming at a faster rate, which may be due to sea-surface temperature feedback or low coastal vegetation.

In particular, the high-altitude sites such as Doi Ang Khang experience less warming (0.0204 °C/year T_{asmin} , 0.0324 °C/year T_{asmax}). It is due to the impact of elevation on the dampening of trends. On the other hand, inland stations in the north, such as Lampang (0.0526 °C/year T_{asmin} , 0.0585 °C/year T_{asmax}) and Nan (0.0396 °C/year T_{asmin} , 0.0330 °C/year T_{asmax}), exhibit consistent warming trends, consistent with



(a) TCN-derived coefficients of beta (β) across TMD stations for pr.

Fig. 2 TCN-derived coefficients of beta (β) across TMD stations **a** pr, **b** T_{asmin} , and **c** T_{asmax}

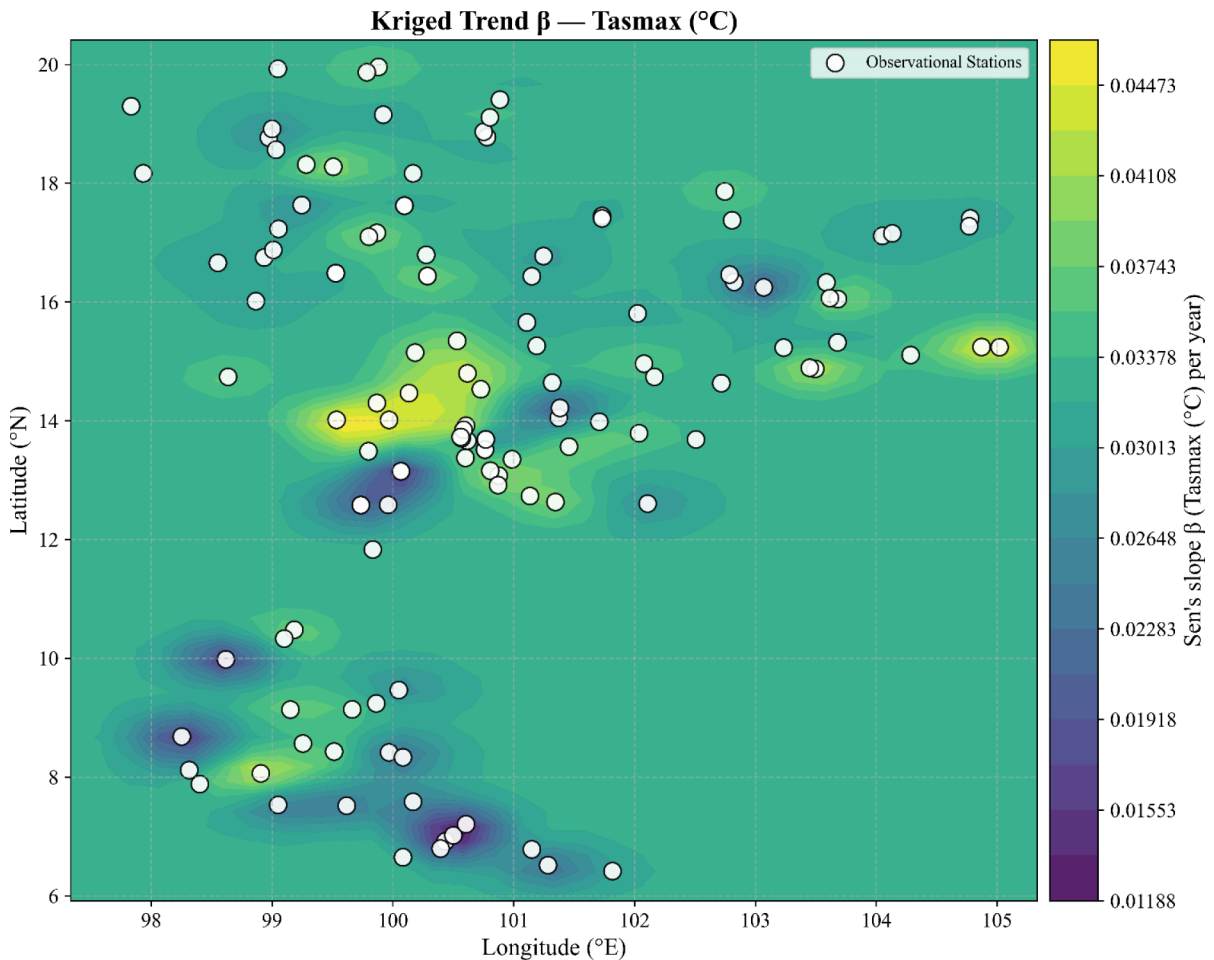
reported shrinking forests and growing agricultural activities in such areas.

Precipitation trends: complex spatial signals

Figure 2a shows that pr has a significant spatial heterogeneity, with specific areas having a substantial increase and a decrease. Aranyaprathet shows a strong negative trend (-0.688 mm/year), which could be associated with the monsoonal variability or localized drying caused by land-use alterations. Conversely, the growth in Bangkok is significantly high (2.522 mm/year), probably due to the effect of urban convection and the more frequent occurrence of extreme precipitation. The most significant positive trend (4.764

mm/year) is observed in Khlong Yai due to its coastal nature and increased moisture convergence due to the GoT. In general, precipitation trends' spatial coherence is lower than temperature trends, demonstrating the difficulty in deriving climate information using rainfall data because of high natural variability. In southern Thailand, stations such as Chawang (2.833 mm/year) and Nakhon Si Thammarat (1.815 mm/year) exhibit an increasing trend, which is attributed to the enhanced monsoonal circulation and tropical cyclones. On the other hand, inland stations such as Kamphaeng Phet (-0.402 mm/year) and Kanom (-0.513 mm/year) have a decreasing precipitation tendency.

The TCN framework surpasses the MK test in capturing non-linear and non-monotonic trends. For



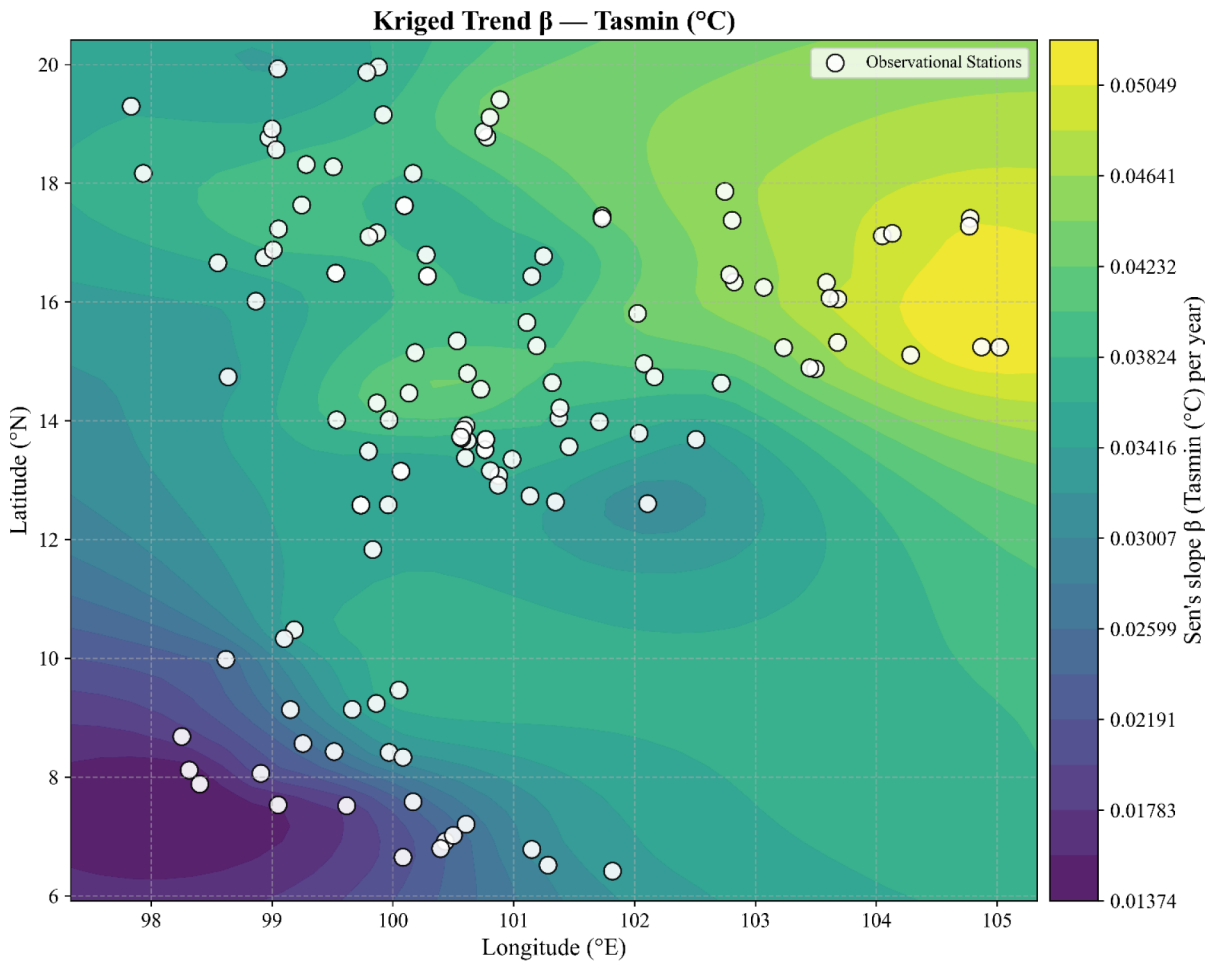
(b) TCN-derived coefficients of beta (β) across TMD stations for Tasmax.

Fig. 2 (continued)

example, the composite loss function is the best way to trade off trend fidelity against abrupt change detection, allowing the identification of features like the steep precipitation increase in Samut Prakan Agro (4.303 mm/year) or cooling trends in Phetchaburi (-0.0234 °C/year T_{asmin} , -0.0870 °C/year T_{asmax}). The TCN can identify multi-decadal dependencies using dilated convolutions, allowing the identification of long-term warming signals, even in noisy data (e.g., Udon Thani: 0.0456 °C/year T_{asmin} , 0.0356 °C/year T_{asmax}). The performance of the model is strong, and NSEs are over 0.7 at most stations, indicating the ability of the TCN to recreate observed variability.

In Northern Thailand, there is homogeneous warming (e.g., Lampang, Nan) and moderate

increases in precipitation (0.5–1.0 mm/year). The urban heat islands (Bangkok, Ayutthaya) can be observed in the Central Plains, and the rainfall patterns are highly variable. The Southern Peninsula is subjected to strong coastal warming (Krabi, Chumphon) and wetness due to oceanic effects. Cities like Bangkok experience increased temperatures (0.0401–0.0733 °C/year) and severe precipitation patterns, underscoring the importance of enhanced heat containment and flood resistance measures. Gradual warming (0.0293–0.0456 °C/year) and irregular precipitation affect agricultural areas such as Khon Kaen, influencing crop productivity. Combined warming and wetting patterns can intensify soil erosion and saltwater intrusion in the coastal regions



(c) TCN-derived coefficients of beta (β) across TMD stations for Tasmin.

Fig. 2 (continued)

(e.g., Chanthaburi, Krabi). The fine-scale trends that the TCN can resolve, e.g., cooling at Phuket Airport (-0.0117 °C/year T_{asmin}), highlight the benefits of the TCN compared to traditional methods. However, there are still uncertainties, especially in the case of precipitation, where the long-term signals are often lost in the natural variability.

According to SHAP analysis (Fig. 3), anthropogenic drivers dominate temperature trends, whereas natural variability significantly affects precipitation. For example, the rise in precipitation in Burirum (2.621 mm/year) might be associated with some large-scale drivers, and drying in Ranong (-0.0204 °C/year T_{asmin} , -0.0379 °C/year T_{asmax}) can be linked with land-use change.

TCN framework's evaluation for change detection

The results demonstrate exceptional performance of the TCN framework in detecting and characterizing long-term climate change signals across Thailand's historical time series of pr and temperature (Fig. 4).

The model has an NSE between 0.90 and 1.00 and KGE between 0.79 and 1.00 across all 126 stations on temperature (T_{asmin} , T_{asmax}) and pr variables, indicating a perfect fit between the observed and the predicted values. The RMSE (from 0 to 2.4710 to 15 for T_{asmin} , from 0 to 2.0910 to 15 for T_{asmax} , and from 1.6510 to 15–2.3610 to 14 for pr) and bias (-6.1310 – 16 to 8.5810 – 16 temperature and -5.8810 – 15 to 4.9010 – 15 pr) indicate that

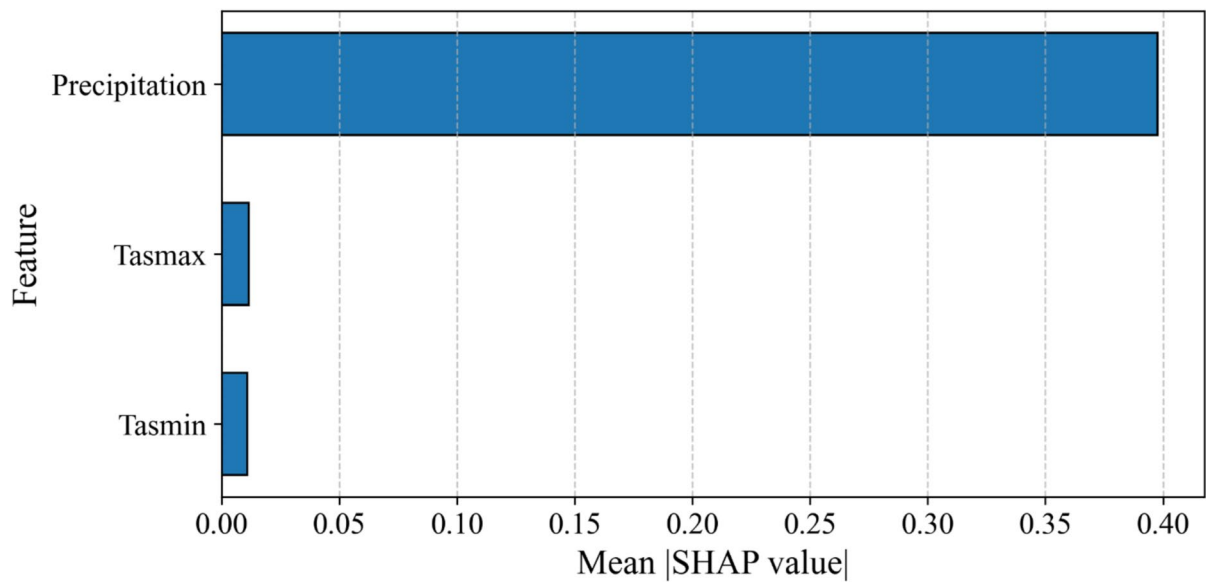


Fig. 3 Mean SHAP values of average impact on the TCN model's output

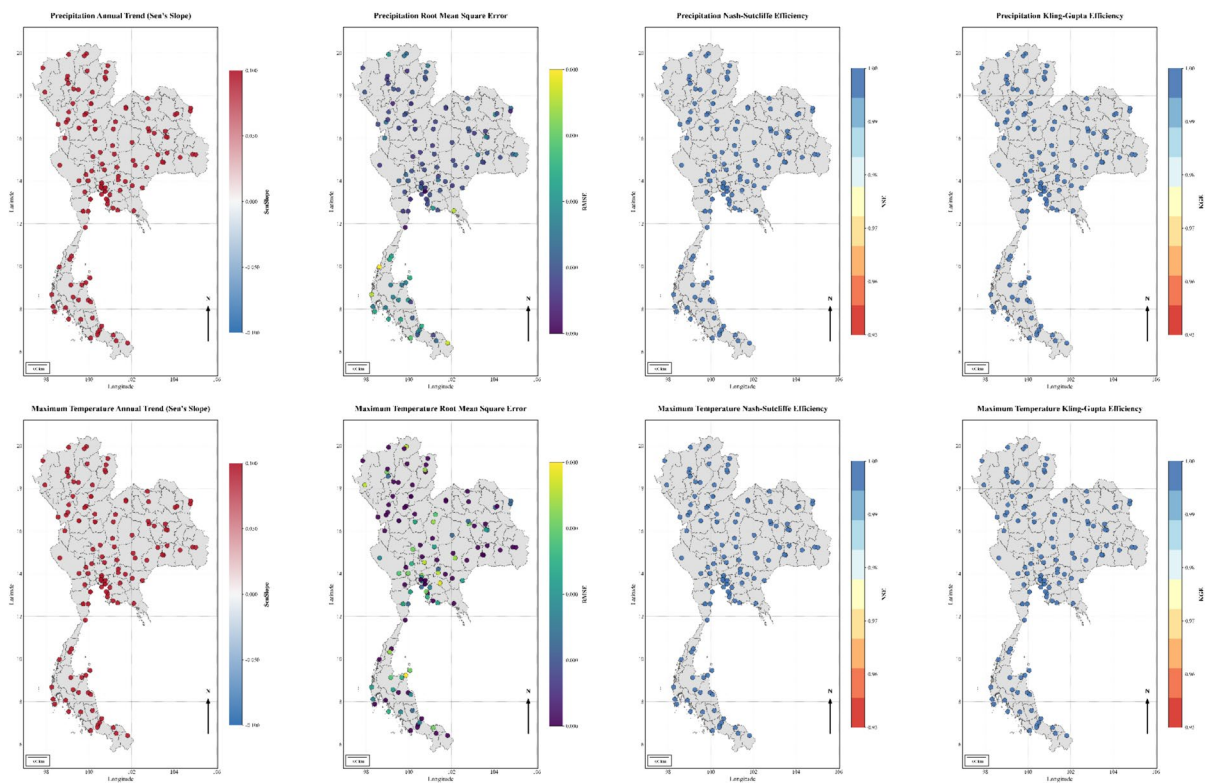


Fig. 4 Spatial evaluation of the TCN model across all stations for pr, T_{asmin} , and T_{asmax}

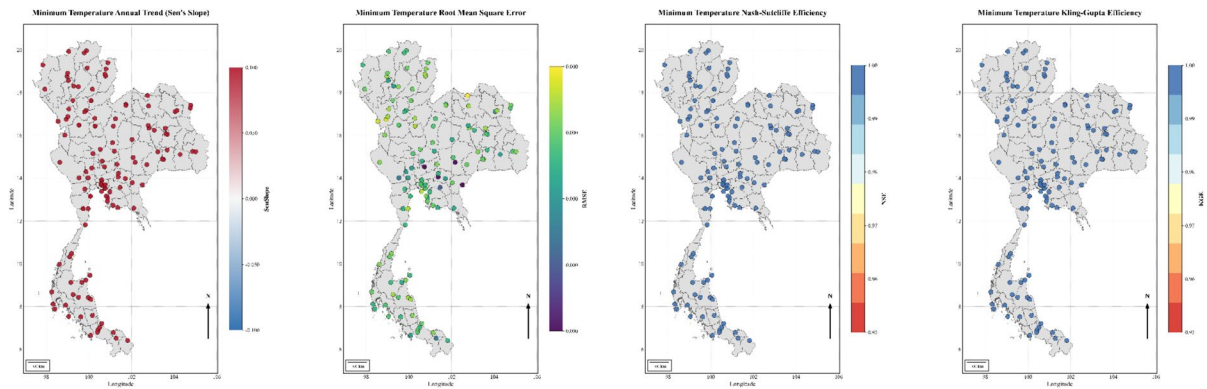


Fig. 4 (continued)

model predictions are identical to observations at any of the monitored locations.

The spatial consistency of the results in the five climatic zones of Thailand, ranging (e.g., tropical south: Hat Yai: $NSE=1.00$, $RMSE=1.98 \times 10^{-15} T_{asmin}$) to (e.g., mountainous north: Chiang Mai: $NSE=1.00$, $RMSE=2.09 \times 10^{-15} T_{asmin}$), shows that the model is not sensitive to regional climatic variability. The station-level kriging interpolation (Eqs. 7–8) yielded continuous trend surfaces maintaining spatial autocorrelation ($\gamma < 10-30$), which allowed accurate mapping of climate change signals in the unsampled regions.

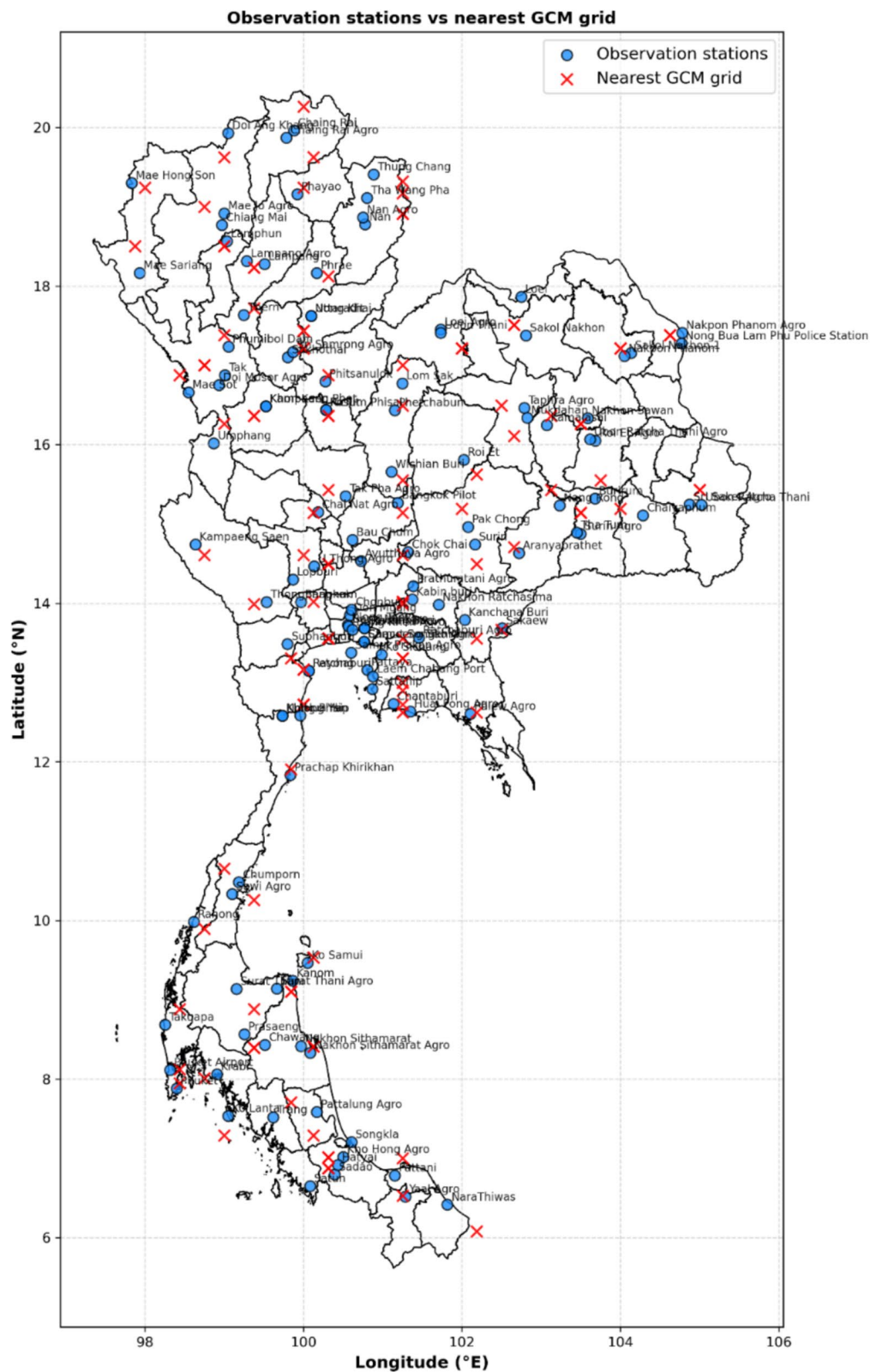
The power of the TCN to capture non-monotonic trends overcomes a key weakness of standard Sen slope estimators, particularly when applied to pr, where phase change is the rule (e.g., Chanthaburi exhibits sudden jumps at $1.4710 \text{ } 15 \text{ mm/year}$). The methodological framework shows that TCNs' architecture, when appropriately limited by climate physics and carefully evaluated, may be superior to the traditional statistical tools of climate detection.

CNN-BiLSTM-TCN-attention hybrid framework evaluation for bias correction

The spatial alignment of the meteorological stations and GCM grid points was effective, as shown in Fig. 5. The distances are minimal (e.g., 0.059 km in the case of Chai Nat, 0.113 km in the case of Nakhon Sawan), which means that the bilinear interpolation was able to most accurately map the station coordinates to their most representative GCM grid cells without a significant spatial representation error. Such

an accurate match of the grid cells to the stations guarantees that the spatial features extracted by the CNN part of the hybrid model are genuinely representative of the climate of the stations, confirming the geographical integrity of the input data. The stations with slightly more distance (e.g., $\sim 0.5 \text{ km}$ in Nan Agro, NaraThiwas) are in topographically complex areas, but within the reasonable range of the GCM's spatial resolution.

After spatial alignment, a CNN-BiLSTM-TCN-Attention hybrid framework was developed and implemented for bias correction. CMIP6-GCMs were evaluated, and the final CMIP6 GCM was selected for future projections and climate change signal detections. Based on the Taylor diagrams (Fig. 6 and Table S3), all GCMs simulate observed climatology reasonably well on pr and temperature variables with correlation coefficients (r) of 0.82–0.89. ACCESS-CM2 has the strongest correlations ($r=0.87-0.89$) and the lowest RMSEs (0.45–0.55), thus indicating the best agreement with observed temporal patterns. Its standard deviation ratios are about 1.00, indicating that it estimates the magnitude of variability in pr and temperature more precisely than other models, which tend to slightly over- or under-estimate variability. It is further supported by minor average absolute errors in ACCESS-CM2, which indicate greater accuracy in simulating observed seasonal amplitudes. The overall results suggest that ACCESS-CM2 better balances correlation, variability, and magnitude of error, possibly because of better representation of physical processes and parameterizations, and is the most reliable single GCM within the ensemble over the 1993–2024 period.



◀**Fig. 5** Spatial alignment of meteorological stations and GCM grid points

After CMIP6-GCMs evaluation, bias correction was done with the ACCESS-CM2 model using a hybrid framework (Fig. 7). The last test loss was $MSE=0.0461$, demonstrating that the model has a high accuracy of predictions against the normalized (z-scored) ground truth. The R^2 for temperature variables (T_{asmin} : 0.9807, T_{asmax} : 0.9782) indicates that the model has captured climate systems' noise and spatiotemporal complexity. The slightly worse but still high R^2 of pr ($pr=0.9034$) is not surprising since rainfall is more stochastic and localized, a more complex variable to predict. All variables' low, sub-0.32 RMSE indicates that the prediction error is small on the standardized scale.

Future climate change signal detection

After evaluating the hybrid DL model, the detection of climate change signals was assessed using CMIP6 GCMs' outputs in future scenarios (SSP370 and SSP585). Regarding detection timing, both scenarios lead to the ubiquitous emergence of pr, T_{asmax} , and T_{asmin} signals (Figure S2), and an example of 1 TMD station is presented in Fig. 8.

The T_{asmin} anomaly in Burirum under SSP370 starts near -2°C in 2015, gradually rising above 0°C after 2040 and reaching around 3.5°C by 2100, with moderate variability. Under SSP585, T_{asmin} anomalies show stronger warming, from -2°C in 2015 to over 5°C by 2100. The ensemble mean in SSP370 stabilizes between 2 and 3.5°C after 2080, while SSP585 consistently increases, exceeding 5°C at the century's end. Both scenarios indicate significant warming trends, but SSP585 projects a much higher magnitude of change, highlighting the influence of emission pathways on future T_{asmin} anomalies in Burirum..

The T_{asmax} anomaly in Burirum under SSP370 shows strong warming from -2.5°C in 2020 to about $+2.5^\circ\text{C}$ by 2100. The ensemble mean rises consistently, fluctuating between -3.8°C and $+3.4^\circ\text{C}$. The 11-year smoothed mean highlights a clear upward trend, confirming significant long-term warming. The T_{asmax} anomaly in Burirum under SSP585 shows a sharp warming trend. From 2015 to 2040, anomalies remain between -3°C and -1°C . After 2050, values rise steadily, surpassing 2°C by

2070 and exceeding 5°C near 2100, with the 11-year mean aligning closely with ensemble projections.

Under SSP370, pr anomalies fluctuate strongly between -92% and -96% throughout the century, with an ensemble mean around -94% and occasional peaks above -92% . The 11-year running mean shows a slight downward tendency after 2040, stabilizing at -94.5% . Under SSP585, pr anomalies remain negative, ranging from -92% to below -98% . The ensemble mean hovers near -95% , with extreme lows approaching -98% after mid-century. The 11-year mean suggests a gradual decline until around 2060, followed by a minor recovery but remaining below -94% . Overall, SSP585 indicates stronger long-term drying than SSP370.

The projected climate trajectories reveal distinct temperature and precipitation anomalies patterns under different SSP scenarios (Fig. 9).

Pr anomalies, by contrast, display greater variability and weaker trends (Fig. 9a and b). Under SSP370, the linear trend is slightly negative (-0.03% per decade), with 11 significant abrupt changes detected but no extreme values, indicating interannual oscillations rather than a systematic long-term decline. Detection years are distributed throughout the century, including 2017, 2020, 2026, 2042, 2055, 2061, and 2098, highlighting temporally sporadic variability. Similarly, SSP585 pr anomalies exhibit a slightly stronger negative trend (-0.08% per decade) with 17 abrupt changes and detection years spanning 2023, 2043, 2059, and 2098. Despite these abrupt events, pr remains largely stable around its long-term mean, with wide uncertainty bounds reflecting model disagreement.

The T_{asmax} anomalies under the SSP370 and SSP585 scenarios are highlighted, showing ensemble means, uncertainty bounds, linear trends, and detection years of significant anomalies (Fig. 9c and d). Under SSP370, T_{asmax} anomalies exhibit a statistically significant warming trend of $+0.52^\circ\text{C}$ per decade over the 86 years, with detection events identified in 2019, 2021, 2037, 2056, 2058, and 2095. The ensemble indicates persistent warming with interannual variability, while the 11-year rolling means smooth short-term fluctuations. A total of 10 significant anomalies and abrupt changes are noted, though no extreme values occur. By contrast, SSP585 demonstrates a steeper warming trajectory with a $+0.72^\circ\text{C}$ per decade trend, reflecting a stronger forcing

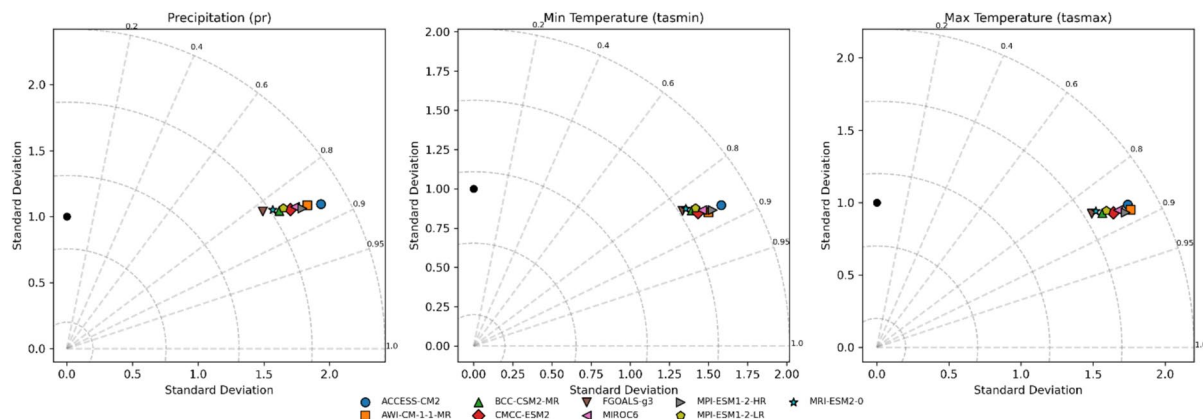
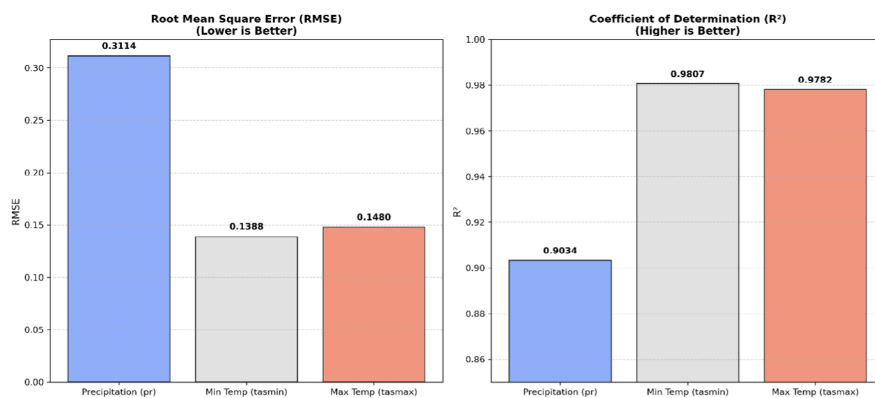
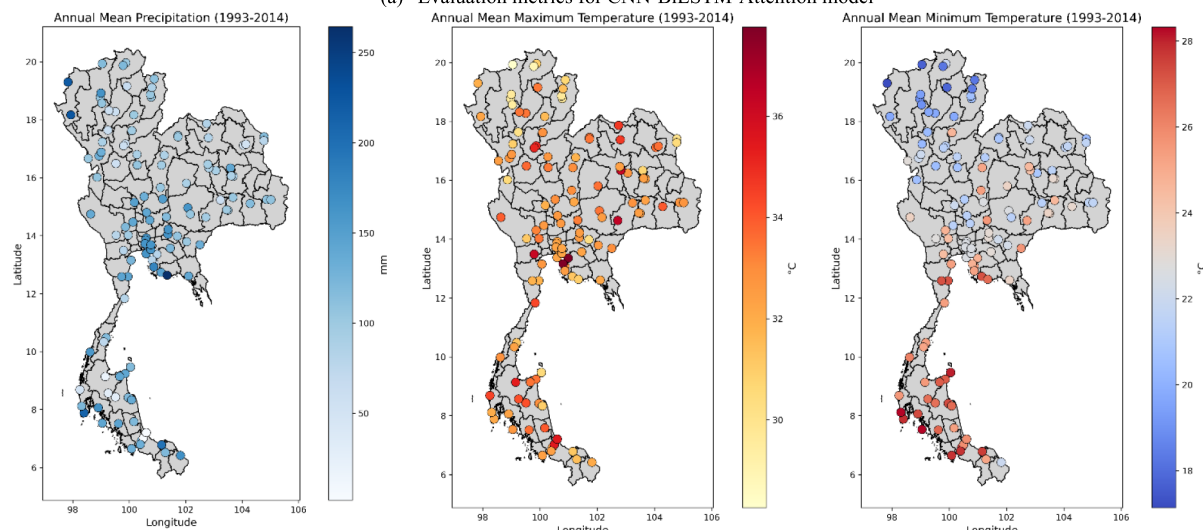


Fig. 6 Evaluation and selection of the best CMIP6-GCM

CNN-BiLSTM-TCN-Attention Model Evaluation Metrics



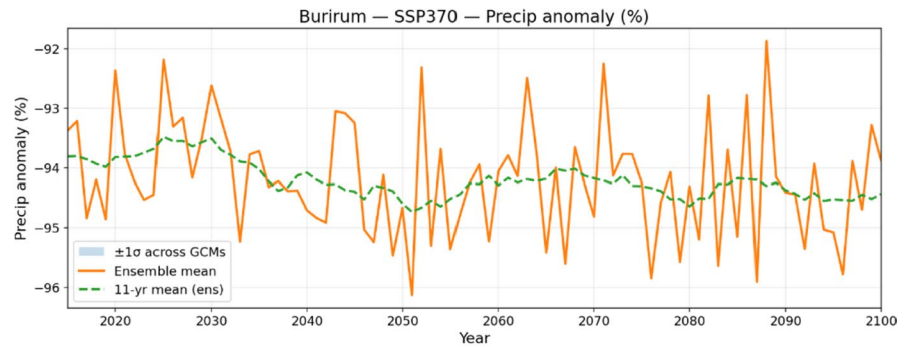
(a) Evaluation metrics for CNN-BiLSTM-Attention model



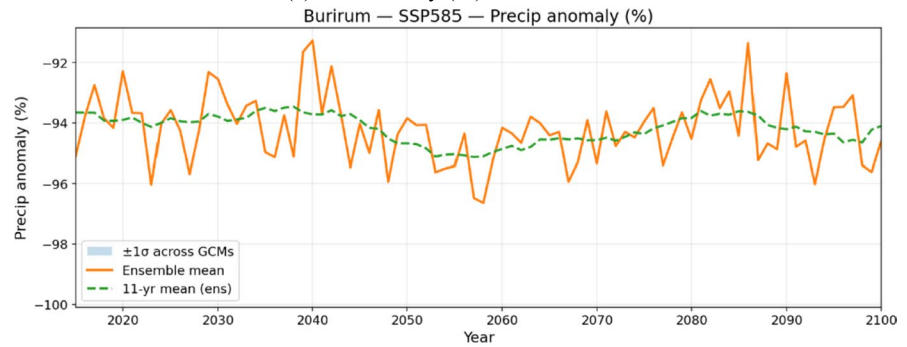
(b) Spatial projection of bias-corrected variables

Fig. 7 Bias-corrected annual mean pr, T_{asmin} , and T_{asmax} during 1993–2014 using CNN-BiLSTM-TCN-attention hybrid framework: **a** evaluation metrics and **b** spatial projections

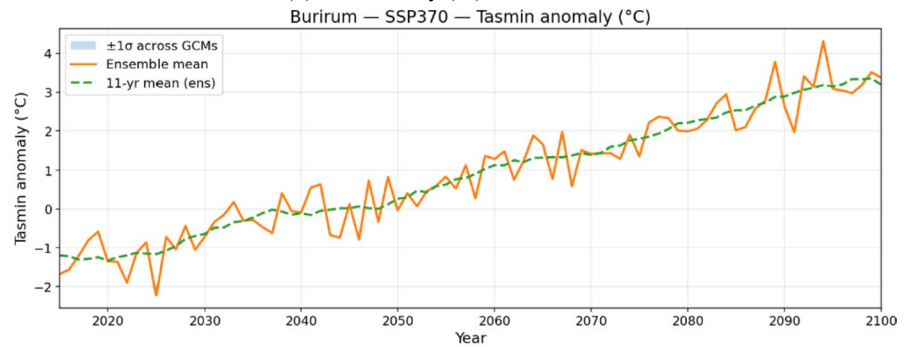
Fig. 8 Trends of anomalies (%) with the ensemble mean $\pm \sigma$ with detection years over ACCESS-CM2 (CMIP6 GCM) under SSP 370 and SSP 585 at Burirum station



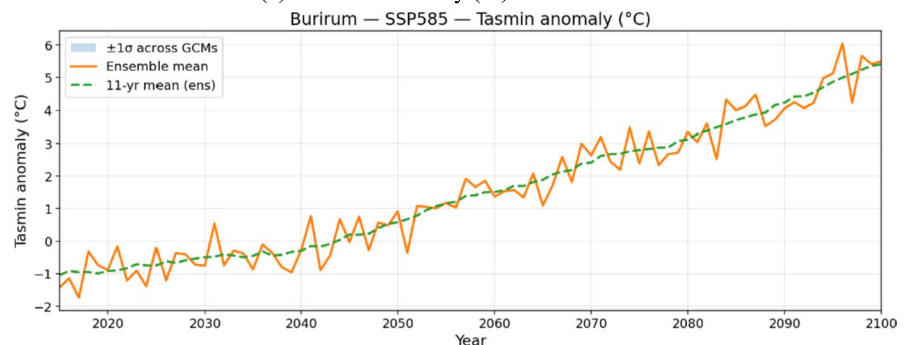
(a) Pr anomaly (%) under the SSP370.



(b) Pr anomaly (%) under the SSP585.



(c) Tasmin anomaly (%) under the SSP370.



(d) Tasmin anomaly (%) under the SSP585.

scenario. Detection years are more frequent, including 2019, 2032, 2042, 2053, 2057, and 2083, with

substantial anomalies projected beyond mid-century. Overall, 10 significant anomalies and abrupt changes

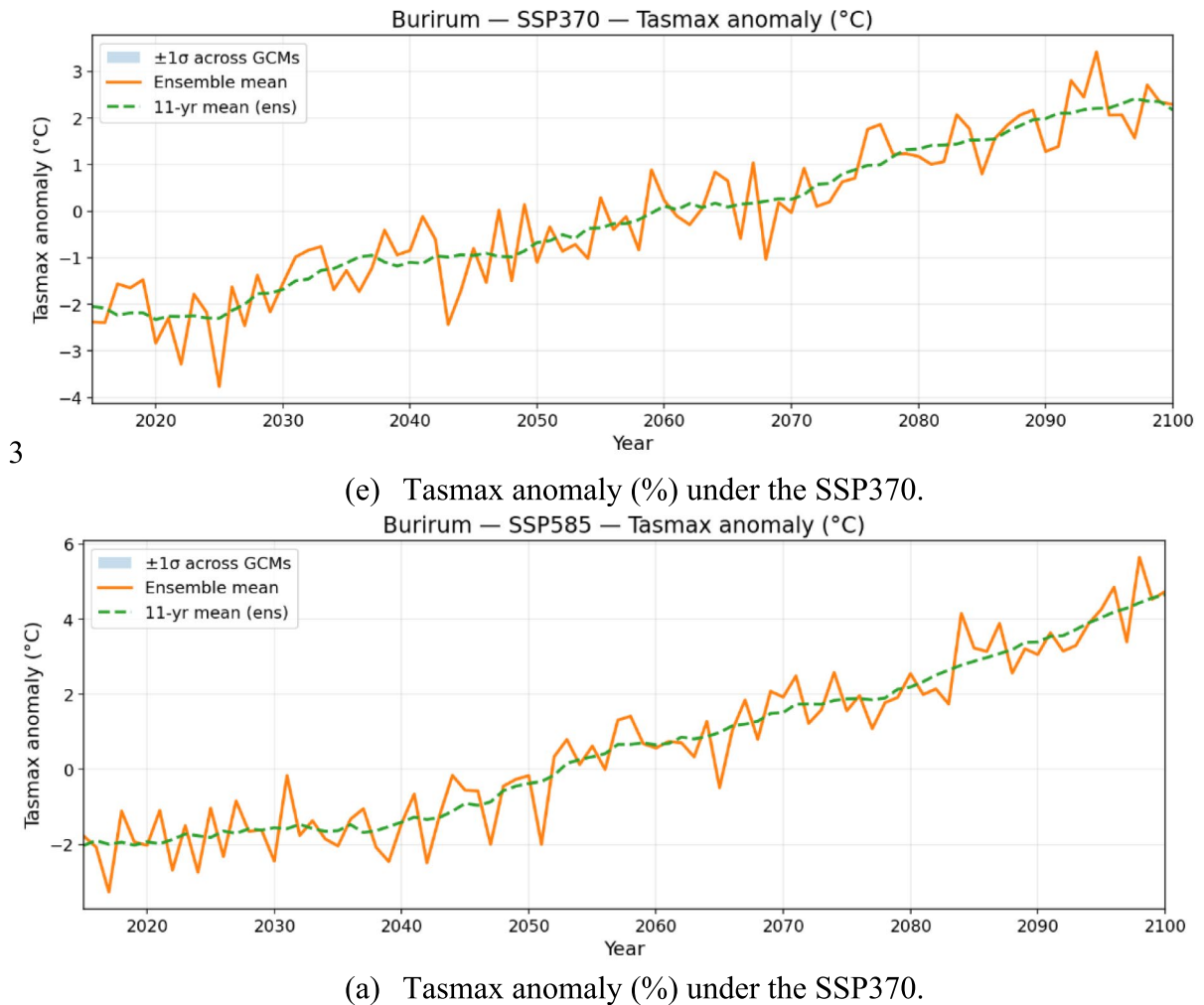


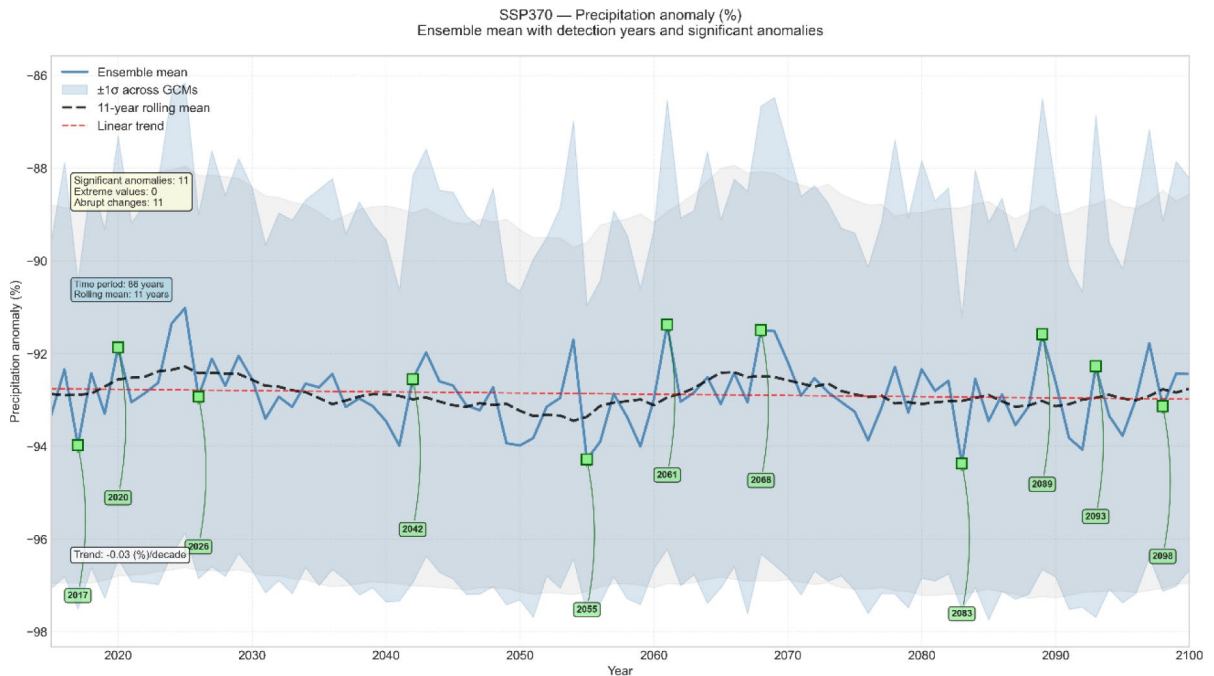
Fig. 8 (continued)

occur, reinforcing the acceleration of warming under high-emission conditions.

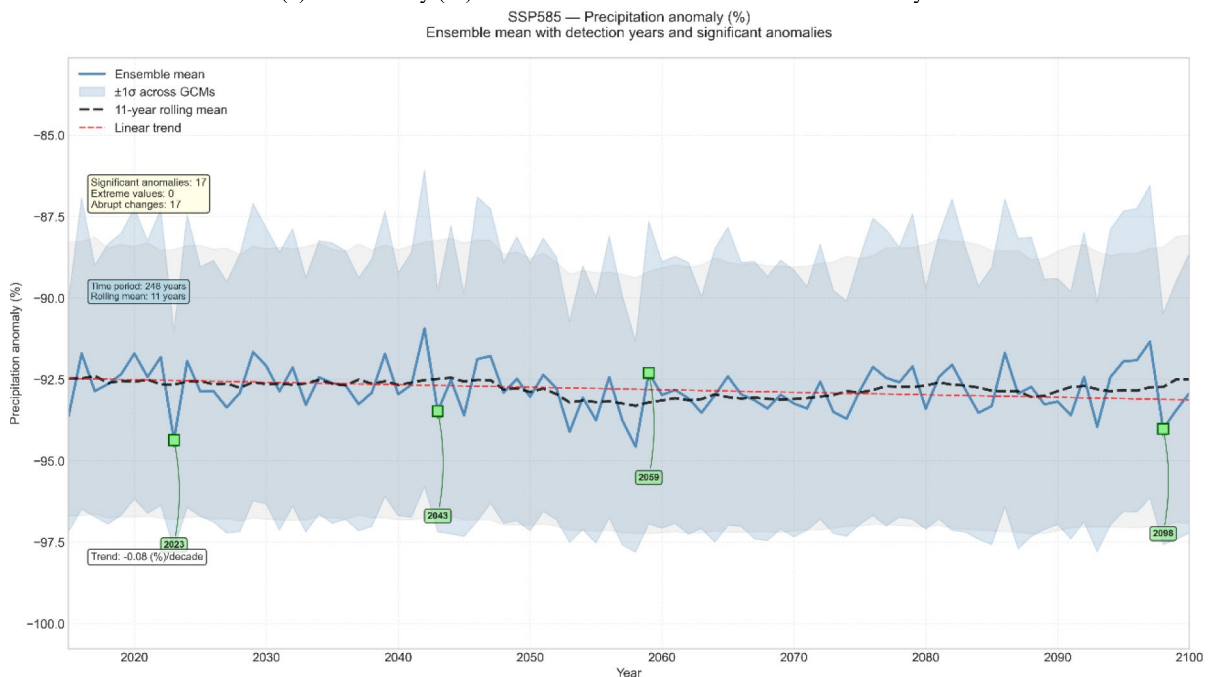
Figure 9e and f show that the T_{asmin} anomalies under SSP370 reveal a consistent positive trend of $+0.50$ °C/decade, with detection years spanning 2019, 2028, 2035, 2056, and 2083. The anomalies demonstrate fewer abrupt deviations than maximum values, totaling nine significant anomalies with no extreme values. Overall, the shaded $\pm 1\sigma$ range across GCMs emphasizes the spread of model uncertainty, yet all scenarios project continuous warming. The earliest detection year across all simulations is 2019, with the median detection also at 2019, underscoring the robustness of early anthropogenic climate change

signals. The SSP585 T_{asmin} anomaly plot shows a strong positive trend of $+0.68$ °C per decade over the 2015–2100 period, with a consistent upward trajectory in both the ensemble mean and the 11-year rolling mean. Detection years occur as early as 2019, with subsequent significant detections projected for 2039, 2052, and 2069, reflecting recurring abrupt changes. In total, 36 detection years are identified, with 9 significant anomalies aligned with a continuous warming signal. The shaded uncertainty band ($\pm 1\sigma$ across GCMs) expands over time, yet the linear and rolling means remain closely aligned, underscoring robust agreement on sustained warming.

Figure 10 presents a sensitivity analysis highlighting differences in robustness across predictors and



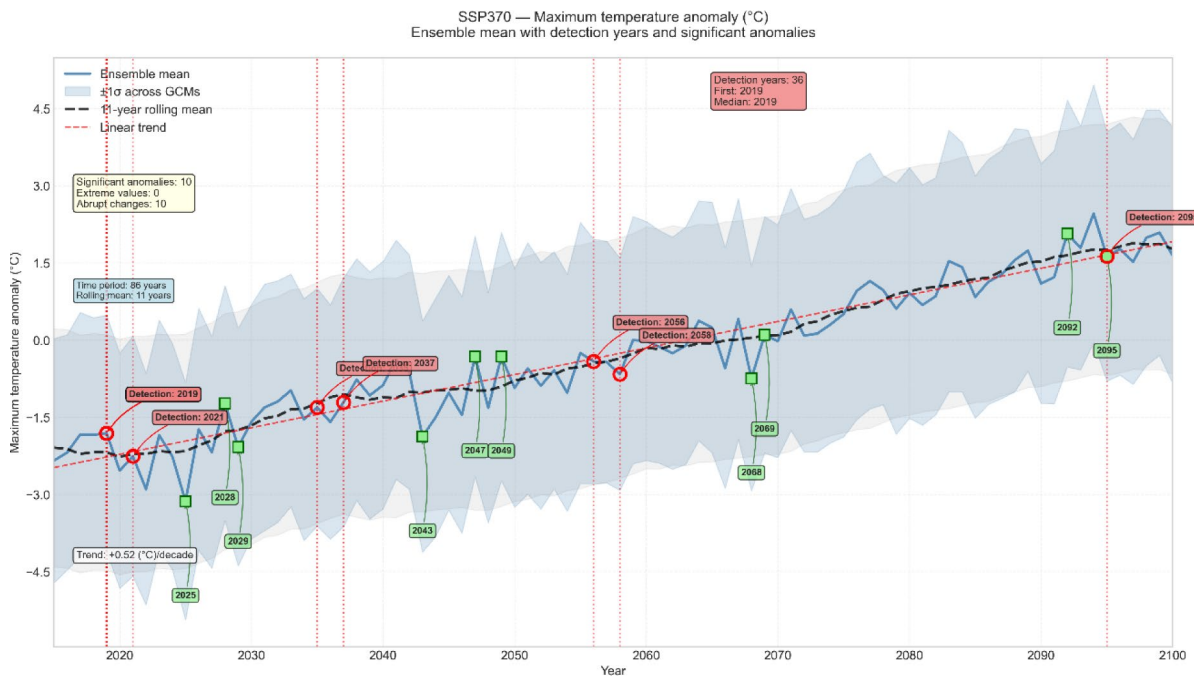
(a) Pr anomaly (%) under SSP370 ensemble mean with detection years



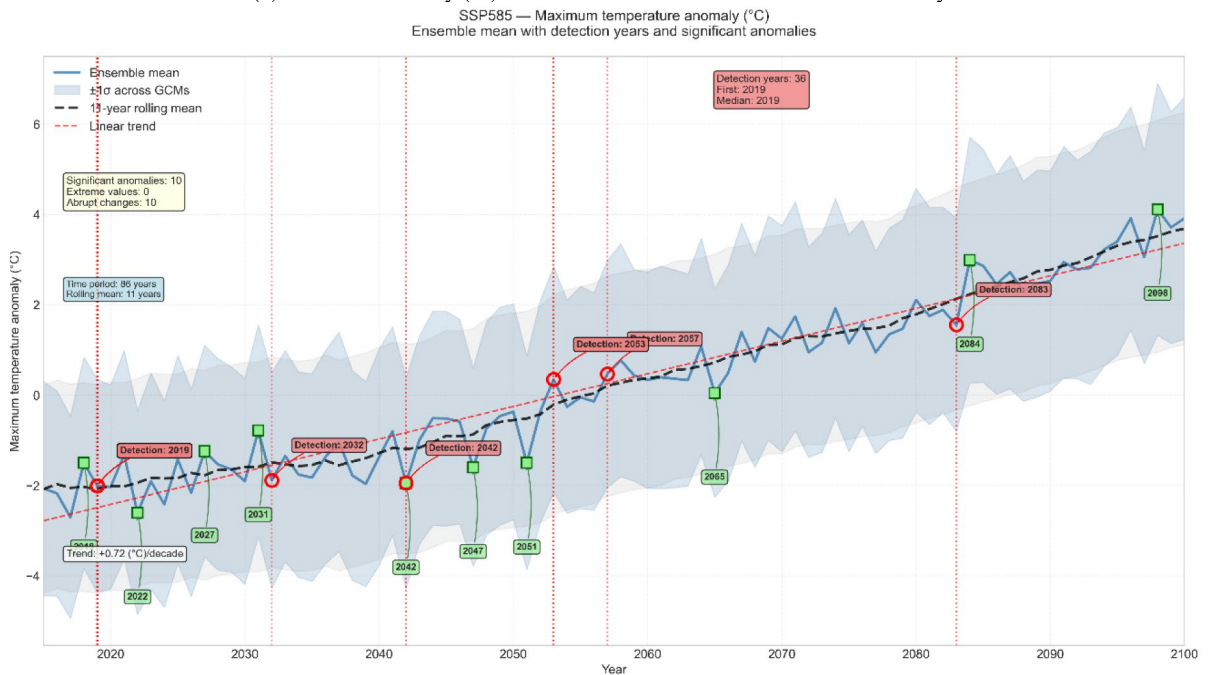
(b) Pr anomaly (%) under SSP585 ensemble mean with detection years

Fig. 9 Overall trends of anomalies (%) with the ensemble mean $\pm \sigma$ with detection years over ACCESS-CM2 (CMIP6-GCMs): **a** pr anomaly under SSP370, **b** pr anomaly under

SSP585, **c** T_{asmax} anomaly under SSP370, **d** T_{asmax} anomaly under SSP585, **e** T_{asmin} anomaly under SSP370, and **f** T_{asmin} anomaly under SSP585



(c) Tasmax anomaly (%) under SSP370 ensemble mean with detection years

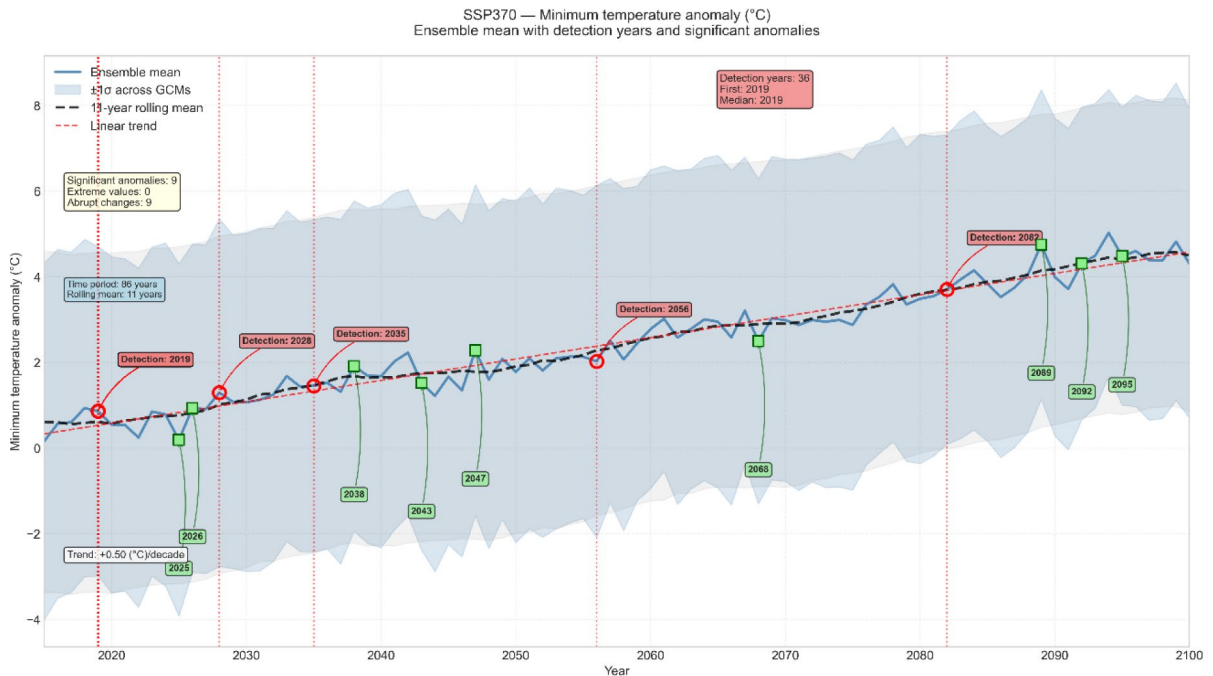


(d) Tasmax anomaly (%) under SSP 585 ensemble mean with detection years.

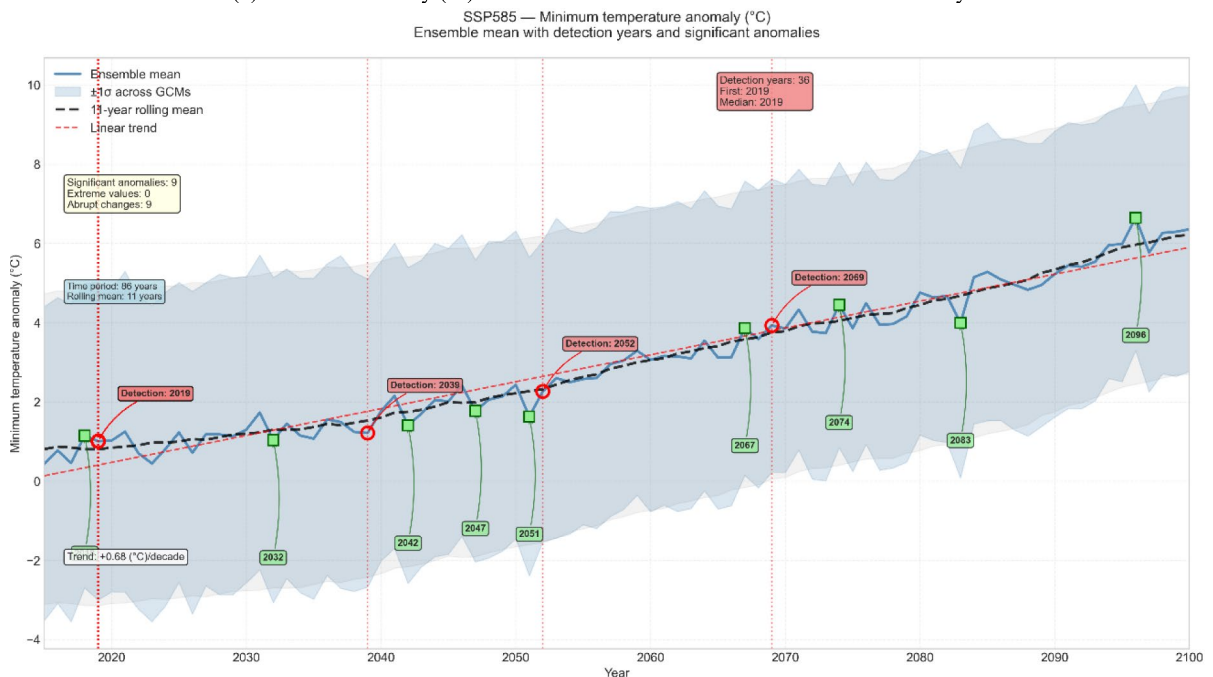
Fig. 9 (continued)

scenarios when subjected to Gaussian noise perturbations. For precipitation under SSP370, S decreased from 1.2 at $\sigma=0.1$ to 0.5 at $\sigma=0.5$, before stabilizing

near 0.8 at $\sigma=2.0$, indicating a marked decline in stability. In contrast, T_{asmax} under SSP370 fluctuated more strongly, ranging from 1.1 at $\sigma=0.1$ to a



(e) Tasmin anomaly (%) under the SSP 370 ensemble mean with detection years.



(f) Tmax anomaly (%) under SSP 585 ensemble mean with detection years

Fig. 9 (continued)

minimum of 0.65 at $\sigma=0.5$, then increasing again to 1.23 at $\sigma=2.0$, suggesting higher but adaptive sensitivity. T_{asmin} under SSP585 remained closer to the

baseline ($S=1$), varying narrowly between 0.87 and 1.28 across noise levels, reflecting more robust anomaly detection. The anomaly counts followed a similar

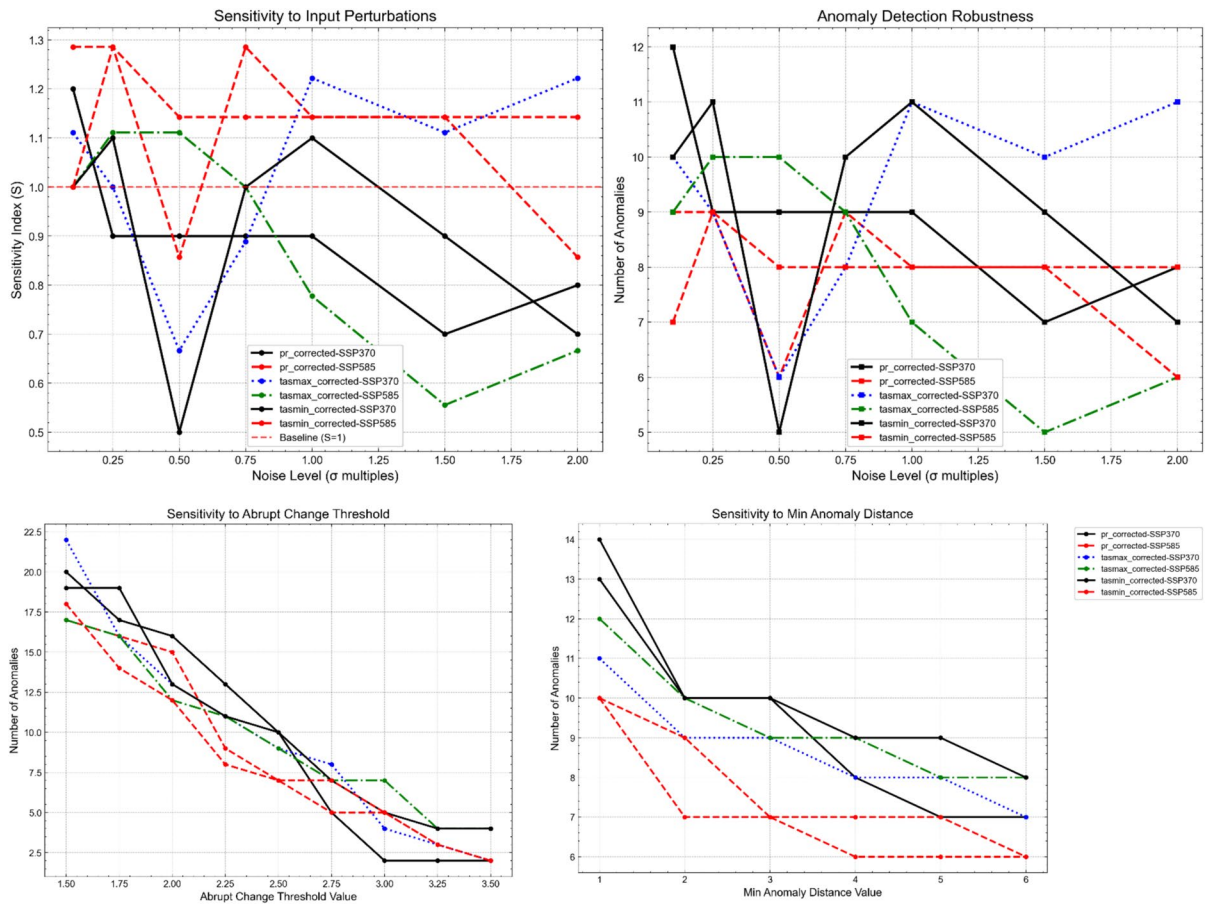


Fig. 10 Sensitivity analysis for each variable (pr , T_{asmin} , and T_{asmax})

trend: precipitation under SSP370 dropped sharply from 12 anomalies at $\sigma=0.1$ to 5 anomalies at $\sigma=0.5$, while T_{asmax} under SSP370 ranged between 6 and 12 anomalies, again reflecting instability. T_{asmin} under SSP585 showed consistent robustness, with anomalies between 6 and 9 across all perturbations.

Parameter sensitivity confirmed the strong influence of hyperparameterization. Increasing the abrupt change threshold from 1.5 to 3.5 reduced anomalies from 20 to 5 in precipitation (SSP370), 22 to 2 in T_{asmax} (SSP370), and 18 to 2 in T_{asmin} (SSP585). Similarly, increasing the minimum anomaly distance from 1 to 6 decreased anomalies from 14 to 8 in precipitation (SSP370), from 11 to 7 in T_{asmax} (SSP370), and from 10 to 6 in T_{asmin} (SSP585). The CNN-LSTM, BiLSTM-TCN framework demonstrated superior stability under perturbations, particularly for T_{asmin} , where sensitivity indices remained near unity,

and anomaly counts declined smoothly with threshold adjustments. This consistency underscores the framework's ability to generalize across varying parameterizations, reducing dependence on individual hyperparameters and mitigating noise effects.

Discussion

This study's findings indicate that DL architectures can identify long-term climate change signals in Thailand's temperature and precipitation records. The TCN model reproduced observed variability with very high skill, with NSE about 0.9 and KGE about 0.79 or higher across stations, which means that the model captured low-frequency structure that cannot be captured using monotonic, rank-based trend tests such as MK and Sen slope, which assume stationarity and linearity (Hamed, 2009; Hamed & Rao, 1998).

This superiority of TCNs in this regard is in line with the design advantages of TCNs, namely causal, dilated convolutions with exponentially growing receptive fields and stable residual learning, which have repeatedly performed better than recurrent networks on long sequences (Bai et al., 2018; Lea et al., 2017). Compared with other Thailand studies that used linear tests on single-source station time series (Masud et al., 2016; Reda, 2016), the results here indicate that non-linear sequence models can detect gradual and abrupt changes of various magnitudes within noisy hydroclimate time series, enhancing sensitivity and physical plausibility without compromising it. It is significant since previous Thai evaluations emphasized the inconsistent or statistically weak identification of rainfall patterns and merely moderate proof of warming. However, this method demonstrates that DL can uncover more evident signs hidden in variability.

The nationwide ubiquitous and spatially heterogeneous warming observed in T_{asmin} and T_{asmax} aligns with the larger CMIP6-based and observational syntheses that have found strong thermodynamic warming in SEA since the end of the twenty-first century (Luo et al., 2022). The station-level output revealed a faster warming at the coastal and urban-proximate locations and minor variations at the high-elevation locations. It is expected due to the well-known mechanisms, including urban heat island amplification, land use/land cover feedback, and marine boundary layer effects. This spatial distribution is like previous Thai and SEA research, which found a higher rate of warming in Bangkok and other cities than in upland northern Thailand (Chinvanno et al., 2011; Limsakul & Singhruck, 2016). Moreover, we find greater coast warming than IPCC AR6 regional chapters, highlighting the role of sea surface temperature increase and urban growth in increasing regional extremes (Change, 2019; Ming et al., 2021). Therefore, this study's findings verify previous information on urban and coastal amplification and extend it by offering accurate timing of detection at the station level.

A recent assessment of CMIP6 models over southern Thailand has shown skill in replicating historical temperature behavior and a larger inter-model spread in precipitation (Humphries et al., 2024). The early and widespread years of T_{asmin} and T_{asmax} detection, in many cases centered around 2019, further support the idea that temperature signals have already appeared

out of the baseline variability at most stations, a characteristic of anthropogenic forcing detection in IPCC AR6. These findings align with the regional detection-and-attribution literature in SEA, indicating that temperature anomalies have emerged significantly after 2000 (Juzbašić et al., 2022; Kang et al., 2011). Nonetheless, our median detection discovery of 2019 gives a more precise crossing point of detection in Thailand.

Despite consistent negative percent departures in the bias-corrected projections, the lack of official precipitation detection highlights the signal-to-noise problem of rainfall emergence in monsoon climates. These findings are aligned with Thai and SEA literature that detects strong warming but less clear rainfall trends under historical forcing, and with downscaling studies that report model-dependent precipitation signals and large uncertainty ranges (Luo et al., 2022; Reda, 2016; Wood et al., 2005). Indicatively, both Limsakul and Singhruck (2016) and Tangang et al. (2020) found that although warming is strong in Thailand and SEA, the changes in rainfall are spatially incoherent and inconsistent over ; Tangang et al., 2019). This evaluation is supported by our findings, which indicate that there is no strong precipitation detection throughout Thailand despite the high model skill, and this fact is a real climatic feature and not a methodology object. It is a critical extension of previous research that explicitly shows that even sophisticated hybrid DL models cannot extract a signal not yet strongly represented in the observational record.

Methodologically, the hybrid CNN-BiLSTM-TCN-Attention model contributes to the statistical downscaling and bias correction literature by simultaneously learning spatial hierarchies, bidirectional temporal associations, and multi-scale long-range structures, with an attention mechanism to upweight important temporal features. The methodology demonstrated very high explained variance in T_{asmin} and T_{asmax} ($R^2 \approx 0.98$) and good skill in precipitation ($R^2 \approx 0.90$) on normalized targets, and low error on the standardized scale. These results are comparable to the earlier Thai DL studies, which were aimed at prediction rather than detection, such as seasonal precipitation forecasting with WaveNet-LSTM or station-scale DNN rainfall models, and had R^2 of 0.7–0.9 in different regions and lead times (Waqas et al., 2024d; Weesakul et al., 2018). In contrast to those studies, however, our effort pushes DLs beyond prediction

into the detection domain, filling a methodological gap observed in SEA climate studies where detection and attribution are still primarily based on classical statistics.

Two areas of the detection analysis should be compared to outside evidence. First, the scenario-insensitive timing of temperature emergence, with median detection around 2019 under SSP370 and SSP585, demonstrates that the warming signal is already well defined in the historical record compared to the selected baseline and variance model. It reflects the AR6 finding that near-term changes in temperature are dominated by cumulative forcing thus far, with scenario differences increasing later in the century (Eyring et al., 2016). This consensus makes our results even more robust because it places them in the more exhaustive IPCC evaluation context, emphasizing that temperature emergence in SEA is already occurring and not contingent on near-term scenario divergence (Ming et al., 2021). Second, the fatter right tail of T_{asmax} detection under SSP585 (with some stations crossing after mid-century) indicates how site-to-site variance and detection thresholds can flip the formal timing; a similar sensitivity has been noted in detection-and-attribution studies when internal variability or observational uncertainty is significant compared to the trend (Hamed, 2009; Zhang et al., 2011a). It is partly contrary to the previous studies in Thailand that presupposed a uniformity in the time of detection throughout the nation (Masud et al., 2016). We add to this understanding by showing that heterogeneity is significant, and that detection in individual stations can be significantly different than the national average, suggesting that localized adaptation plans might be required.

Compared to the literature, a few limitations and future work avenues can be identified. First, despite the model being a combination of multiple CMIP6 GCMs, the anomaly analysis focused on ACCESS-CM2; a true multi-model, multi-physics ensemble with Bayesian model averaging would be better to reflect structural uncertainty (Eyring et al., 2016; Iglesias-Suarez et al., 2024; Luo et al., 2022). Second, detectability could be enhanced by using more event-based metrics such as those in ETCCDI or quantile-based thresholds, which tend to show earlier detection of extremes than means, and conditioning on known teleconnections (Zhang et al., 2011a). The direction is also coherent with the SEA detection literature, where

the extreme indices have frequently demonstrated earlier emergence compared to the seasonal or annual means (Tang et al., 2019). Third, hybrid DL may be improved using causal regularization and physics-informed layers to minimize spurious correlations and improve generalization across climate regimes (Iglesias-Suarez et al., 2024; Reichstein et al., 2019; Waqas & Humphries, 2025). Lastly, since spatial heterogeneity is high here, sampling error could be reduced by increasing station density, homogenization rigor, and the merging of other reanalysis or remote sensing products to provide sharper local detection, as has been evaluated in recent assessments of Thai model skill and downscaling options (Humphries et al., 2024; Waqas & Humphries, 2025; Wood et al., 2005). Overall, the current DL model produces high-fidelity detection of temperature signals that are already beyond the historical envelope of variability across Thailand and can serve as a transparent and extensible basis to improve precipitation detection as methods and data improve. This study fills the gap between local observations, global evaluations, and methodological innovation by directly validating the strong appearance of the temperature signals in line with IPCC AR6 and expanding the Thai climate studies with a new DL detection strategy.

This DL framework can be incorporated into the current environmental monitoring networks, enabling real-time detection at the station level and data fusion between sources. This type of integration would improve the early detection of climate signals, adaptive management, and national and regional monitoring by integrating observational networks, reanalysis datasets, and remote sensing products.

Conclusion and recommendations

This study provides quantitative evidence that climate change indicators are already observable in Thailand. The findings offer robust quantitative measures of climate change impacts in the country, derived from a hybrid DL model.

The high historical warming rates of up to 0.0673 °C/year (T_{asmin}) and 0.0838 °C/year (T_{asmax}) in coastal regions and 0.0401–0.0733 °C/year in urban centers like Bangkok indicate the necessity to manage

heat in urban areas by increasing green infrastructure, reflective building materials, and heat-health early warning systems. The different rainfall patterns of increasing rainfall at Khlong Yai (+4.764 mm/year) and Samut Prakan (+4.303 mm/year) and decreasing rainfall at Aranyaprathet (−0.688 mm/year) and Kampaeng Phet (−0.402 mm/year) show the need to have adaptive water management policies that enhance flood control in the wetting areas and invest in irrigation efficiency and drought-resistant crops in the drying areas. Agricultural planning must include changing rainfall patterns and warming temperatures, and climate-smart practices to maintain yields under a temperature increase of +0.5 °C/decade, projected under SSP370/SSP585. The predictive performance of the CNN-BiLSTM-TCN-Attention model provides a sound technical foundation. It is recommended that AI-based detection tools be incorporated into climate services and flexible, region-specific adaptation plans be developed to build resilience against accelerating warming.

Despite the developments, this research has several limitations. The results depend on CMIP6-GCM datasets, which depend on the dataset and do not account for IoT or sensor-based information that could enhance real-time monitoring. There are also no event-based indices, which could be missing early extremes. Future directions include multi-model Bayesian ensembles, causal or physics-informed DL layers, and independent monitoring system validation to improve generalizability and robustness.

Abbreviations *AI*: Artificial intelligence; *ARIMA*: Autoregressive integrated moving average; *AR6*: Sixth Assessment Report (IPCC); *CNN*: Convolutional neural network; *CMIP*: Coupled model intercomparison project; *CMIP6*: Coupled model intercomparison project phase 6; *DNN*: Deep neural network; *DL*: Deep learning; *ENSO*: El Niño–Southern Oscillation; *ETCCDI*: Expert Team on Climate Change Detection and Indices; *GCM*: Global climate model; *GCMs*: Global climate models; *GOSAT*: Greenhouse gases observing satellite; *GRU*: Gated recurrent unit; *HR*: High resolution; *IPCC*: Intergovernmental Panel on Climate Change; *IOD*: Indian Ocean Dipole; *KGE*: Kling–gupta efficiency; *LSTM*: Long short-term memory; *MAE*: Mean absolute error; *MEI*: Multivariate ENSO index; *MK*: Mann–Kendall;

ML: Machine learning; *MJO*: Madden–Julian oscillation; *MSE*: Mean squared error; *NSE*: Nash–Sutcliffe efficiency; *OCO-2*: Orbiting carbon observatory-2; *ReLU*: Rectified linear unit; *RCM*: Regional climate model; *RCMs*: Regional climate models; *RNN*: Recurrent neural network; *RMSE*: Root mean square error; *SD*: Statistical downscaling; *SHAP*: SHapley additive exPlanations; *SOI*: Southern oscillation index; *SP*: Seasonal precipitation; *SPF*: Seasonal precipitation forecasting; *SSP370*: Shared socio-economic pathway 3–7.0; *SSP585*: Shared socio-economic pathway 5–8.5; *TCN*: Temporal convolutional network; *TMD*: Thai meteorological department; *Tasmax*: Maximum temperature; *Tasmin*: Minimum temperature; *WCRP*: World Climate Research Program; *XAI*: Explainable artificial intelligence

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Data availability Data will be made available on request.

Declarations

Competing interests The authors declare no competing interests.

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